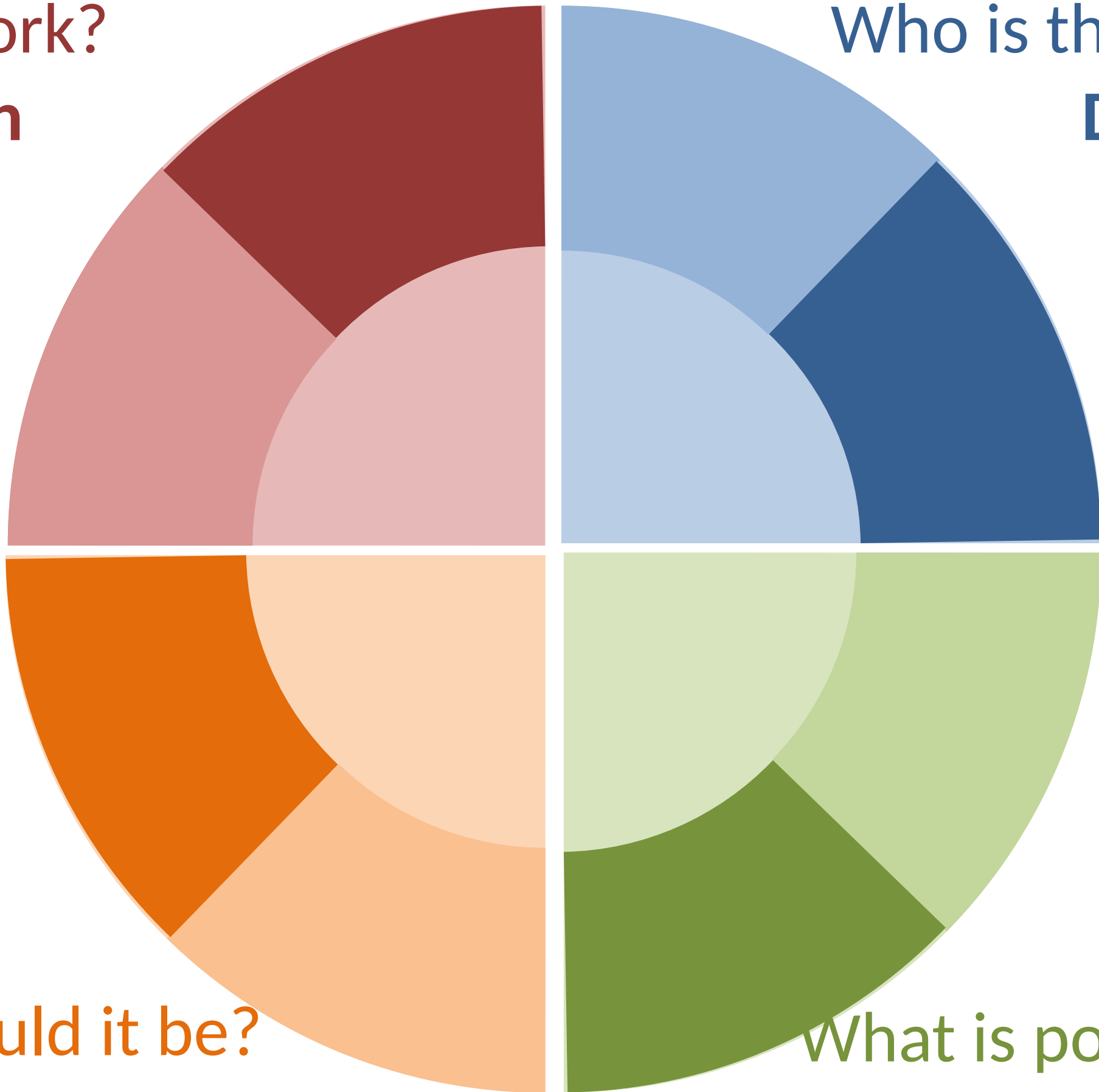


Does it work?
Evaluation

Who is the user?
Discover

What should it be?
Design

What is possible?
Invention

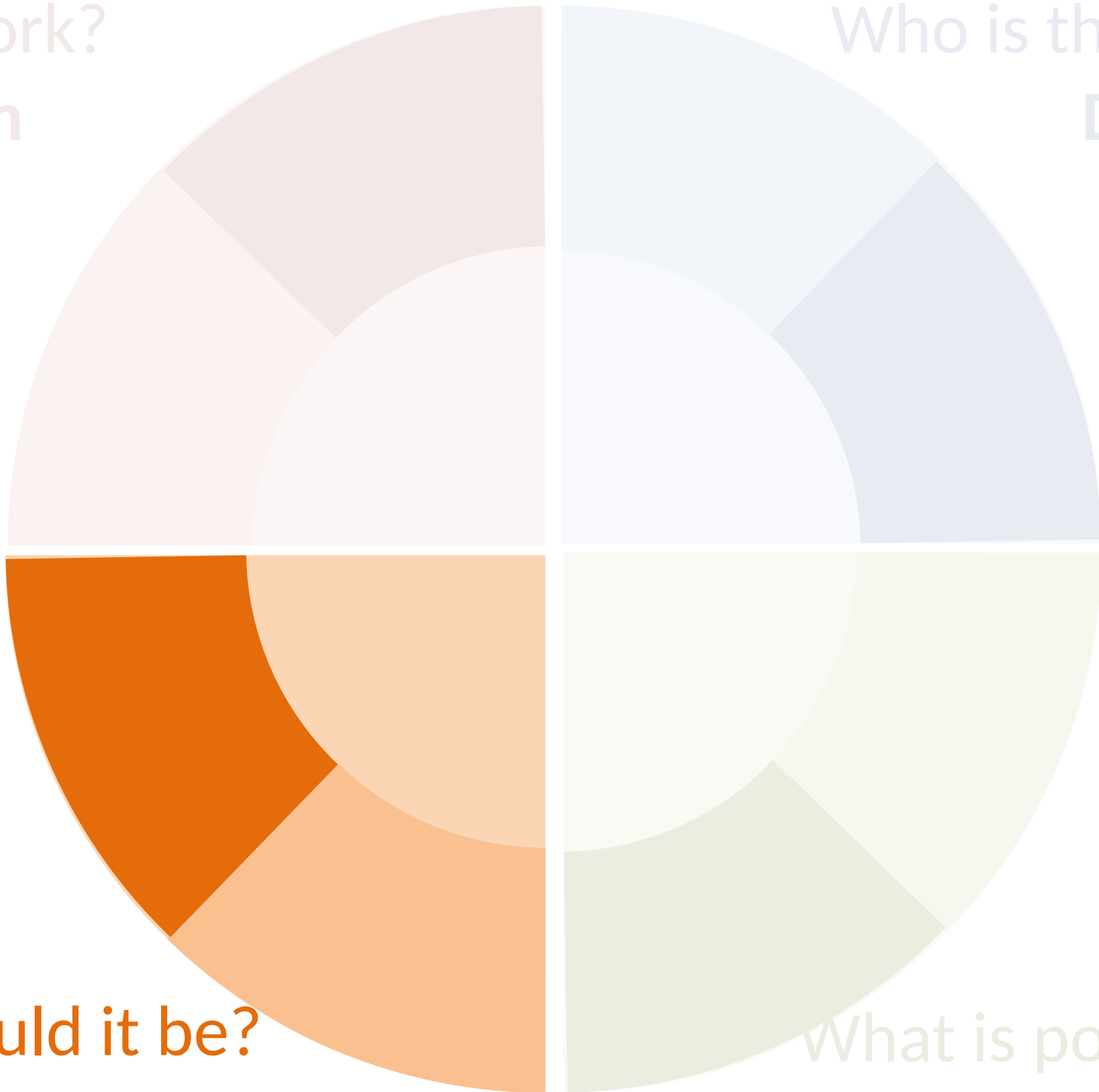


Does it work?
Evaluation

Who is the user?
Discover

What should it be?
Design

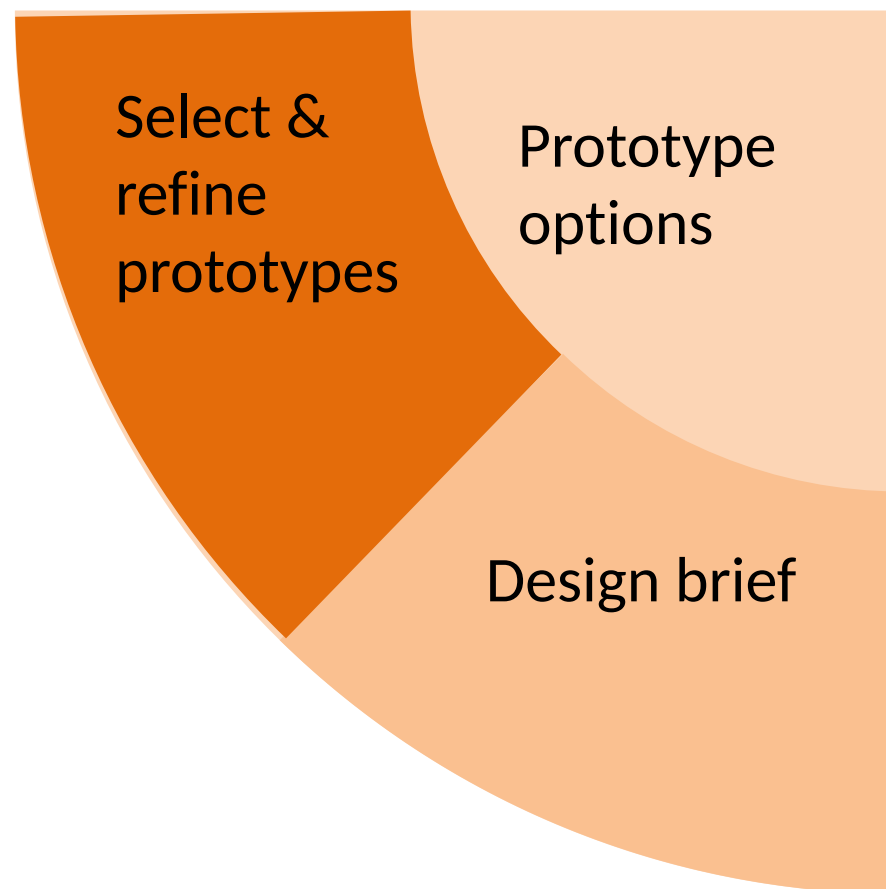
What is possible?
Invention



Design Resources

- Low-fidelity prototype
- Medium-fidelity prototype
- High-fidelity prototype

Analyze information



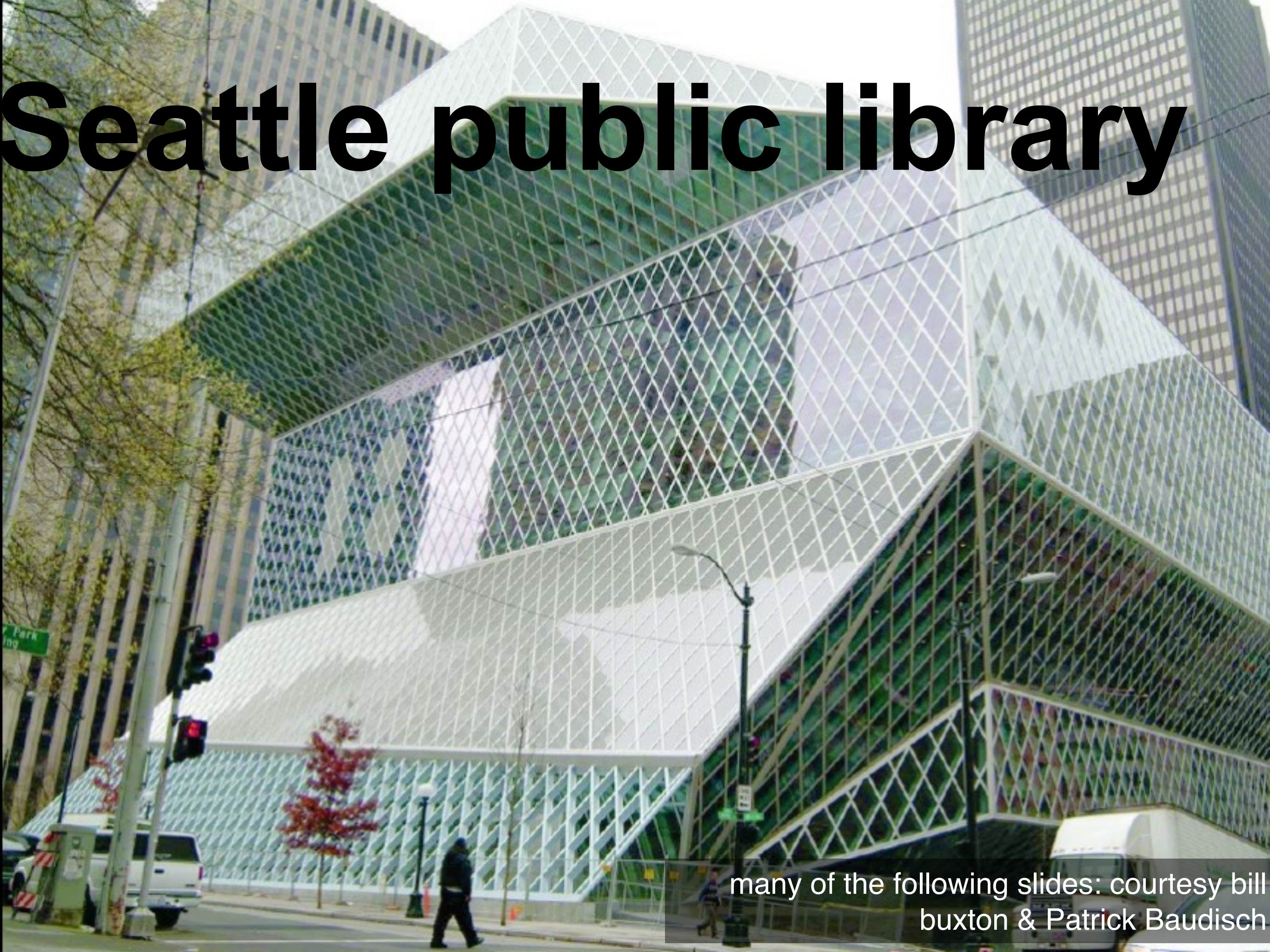
Collect information

- sketches

What should it be?

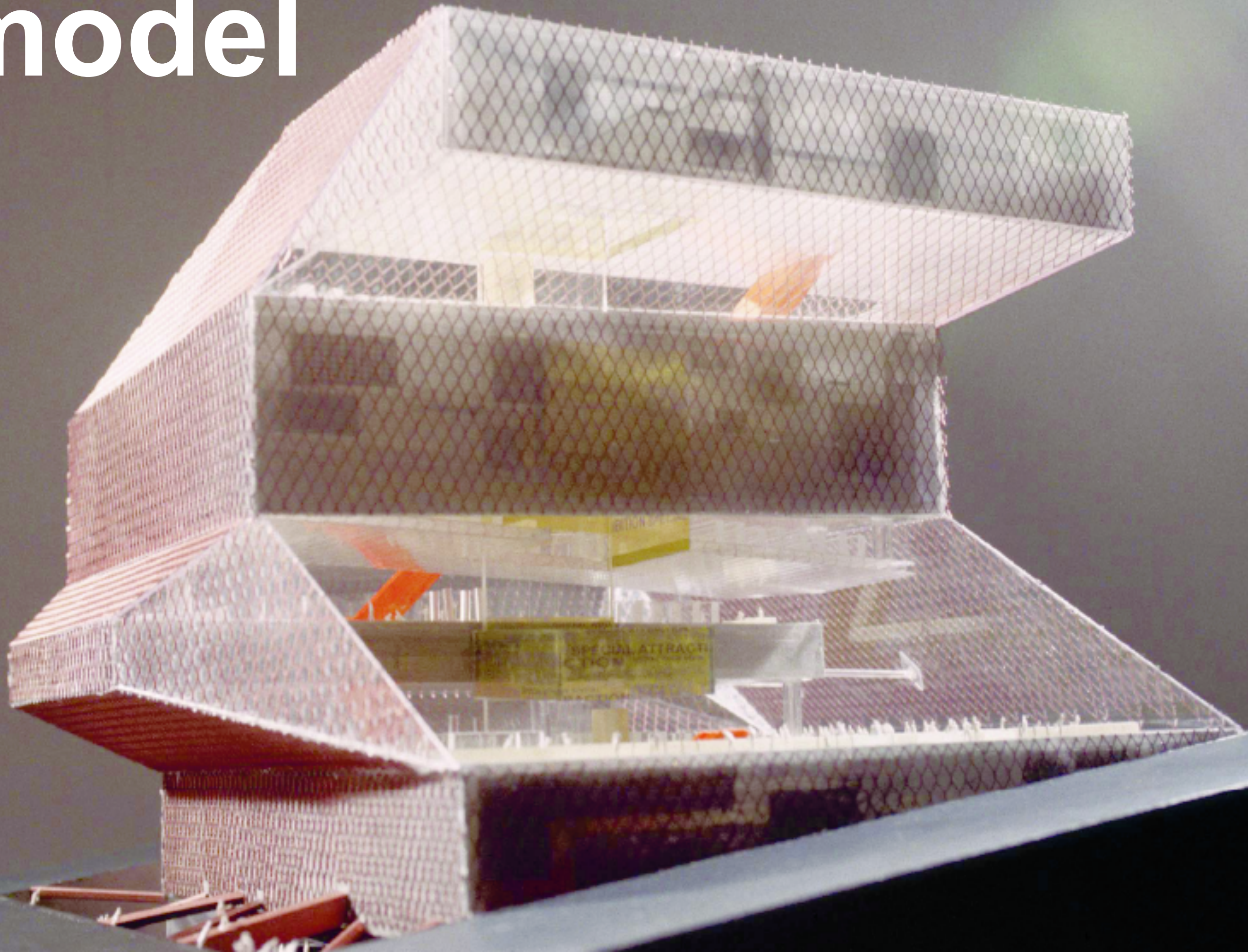
Design

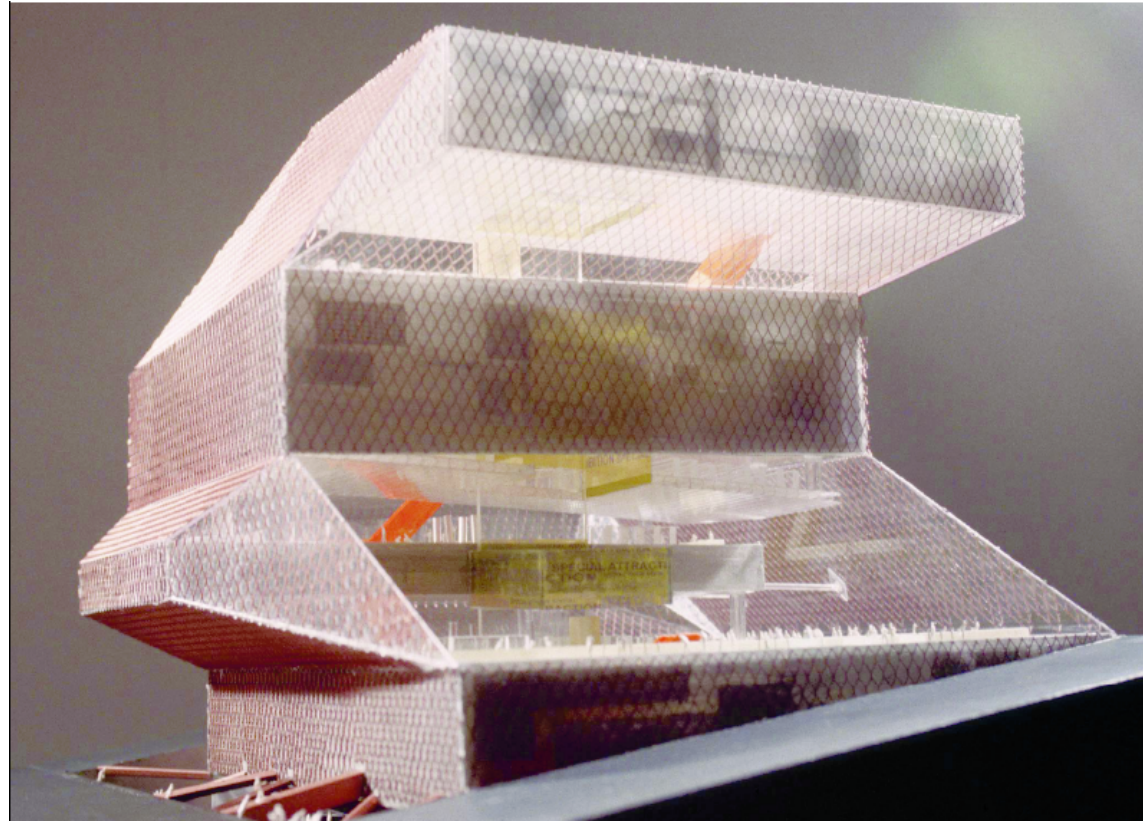
Seattle public library



many of the following slides: courtesy bill
buxton & Patrick Baudisch

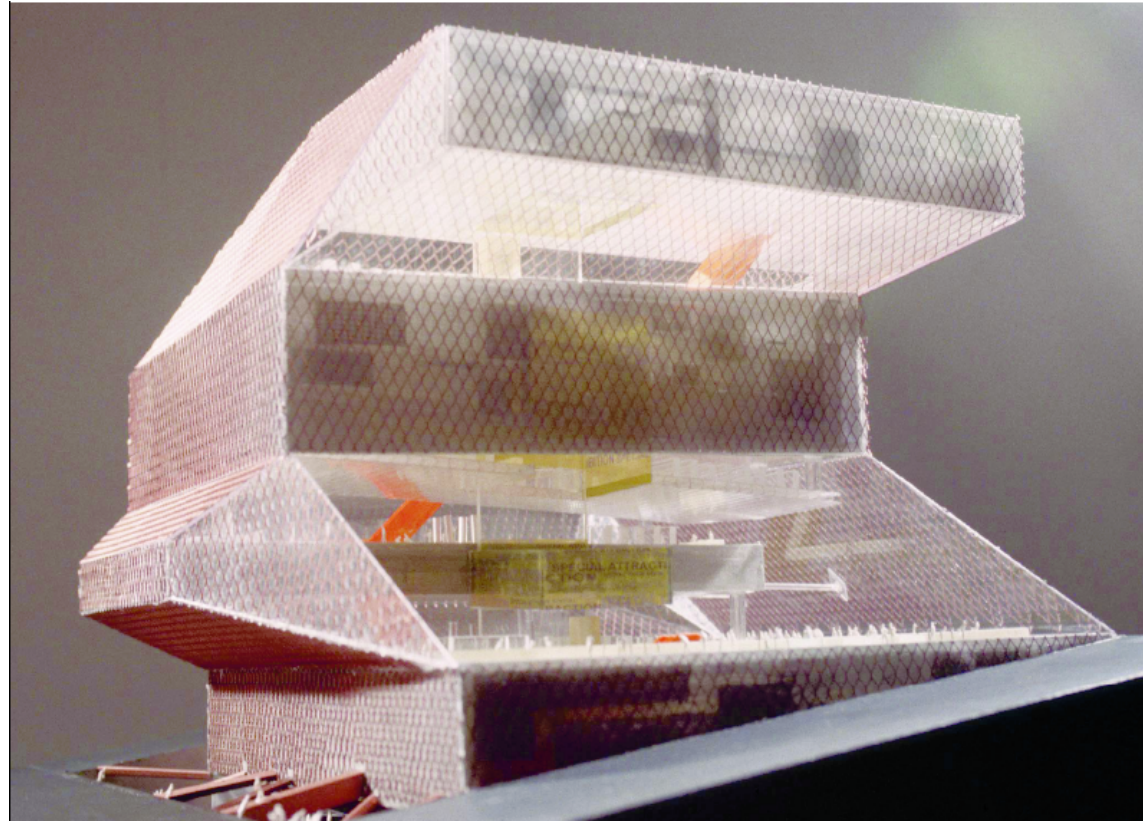
model





model
dec1999



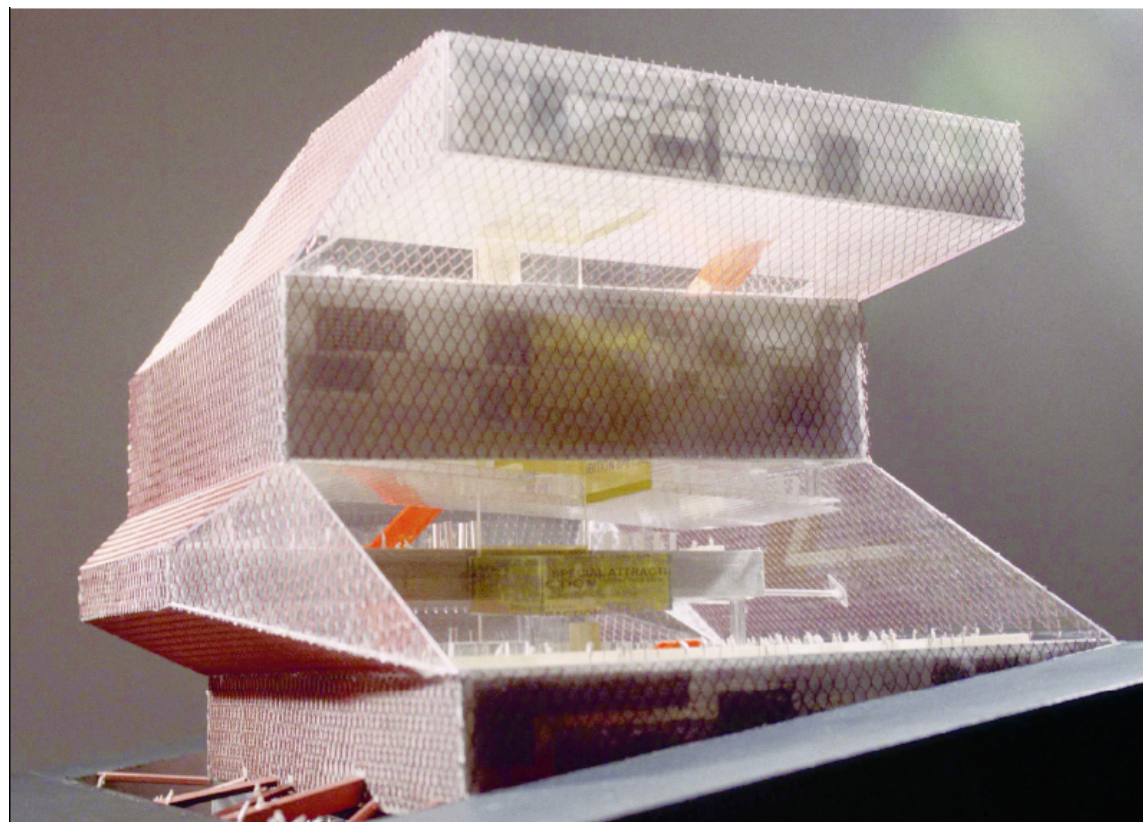


model
dec 1999

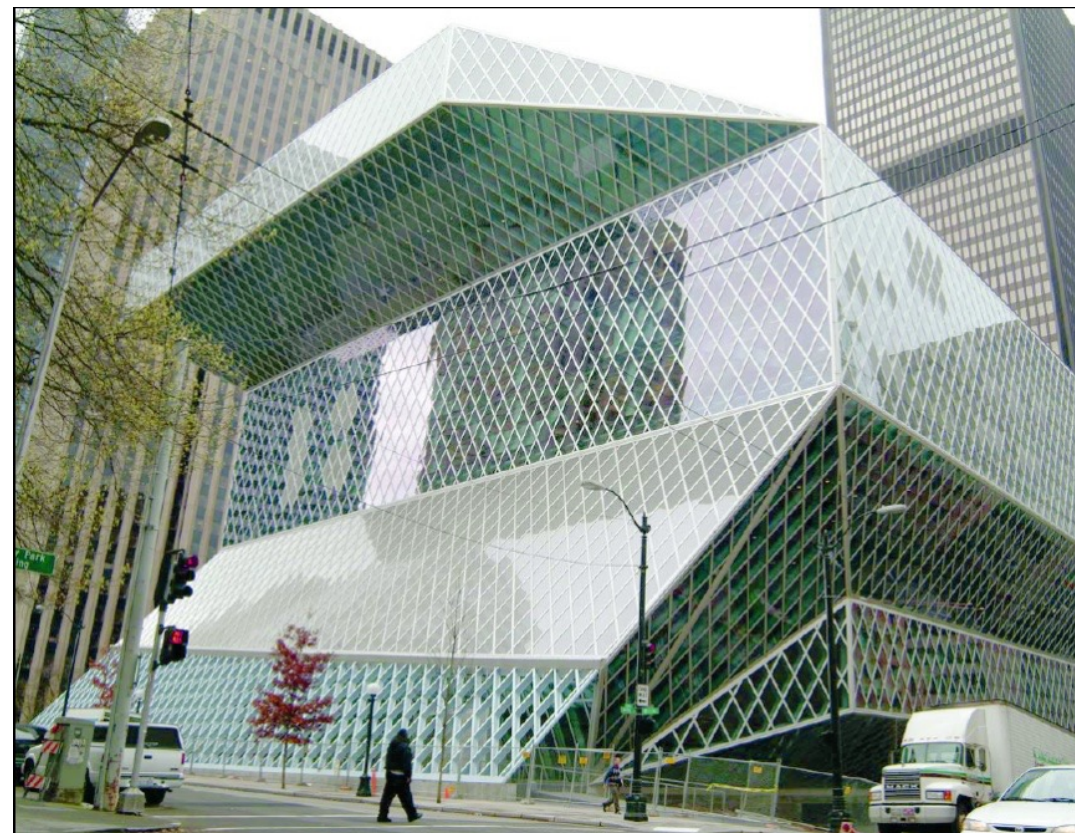


open
jul 2004

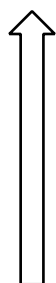




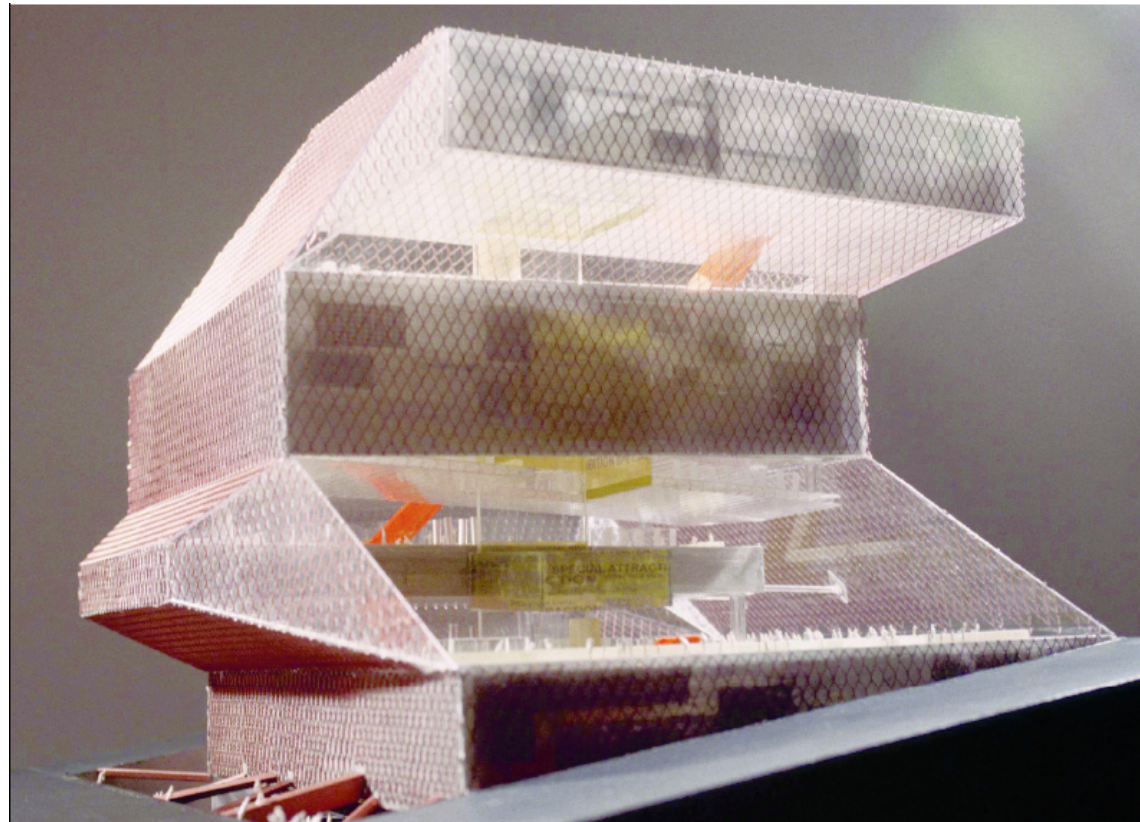
model
dec 1999



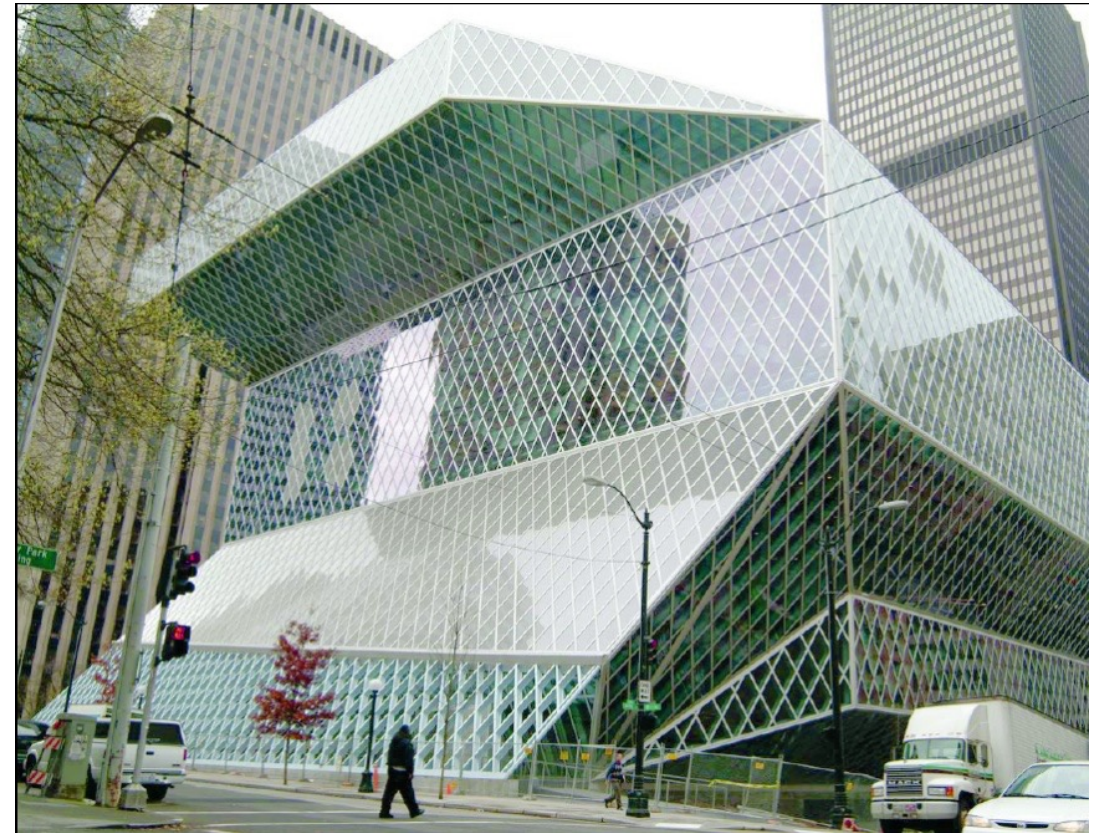
open
jul 2004



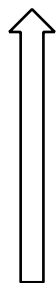
call
nov 1998



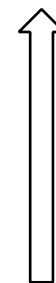
model
dec 1999



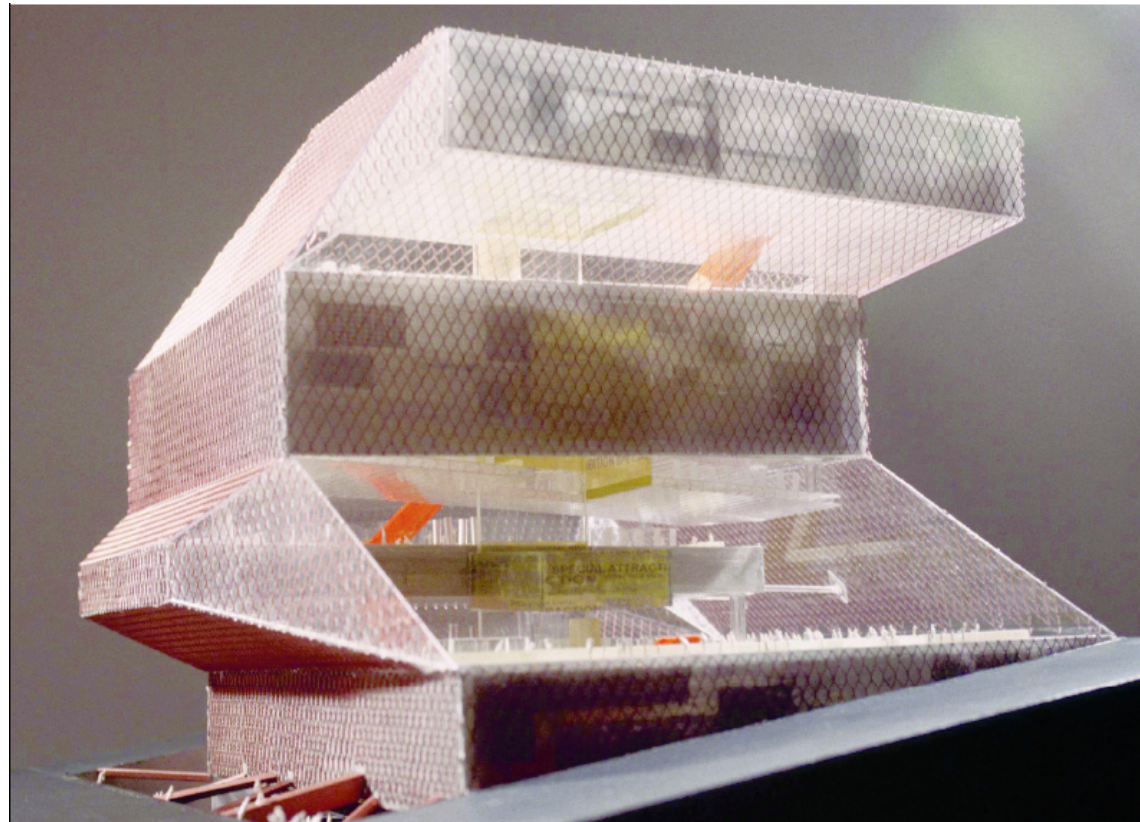
open
jul 2004



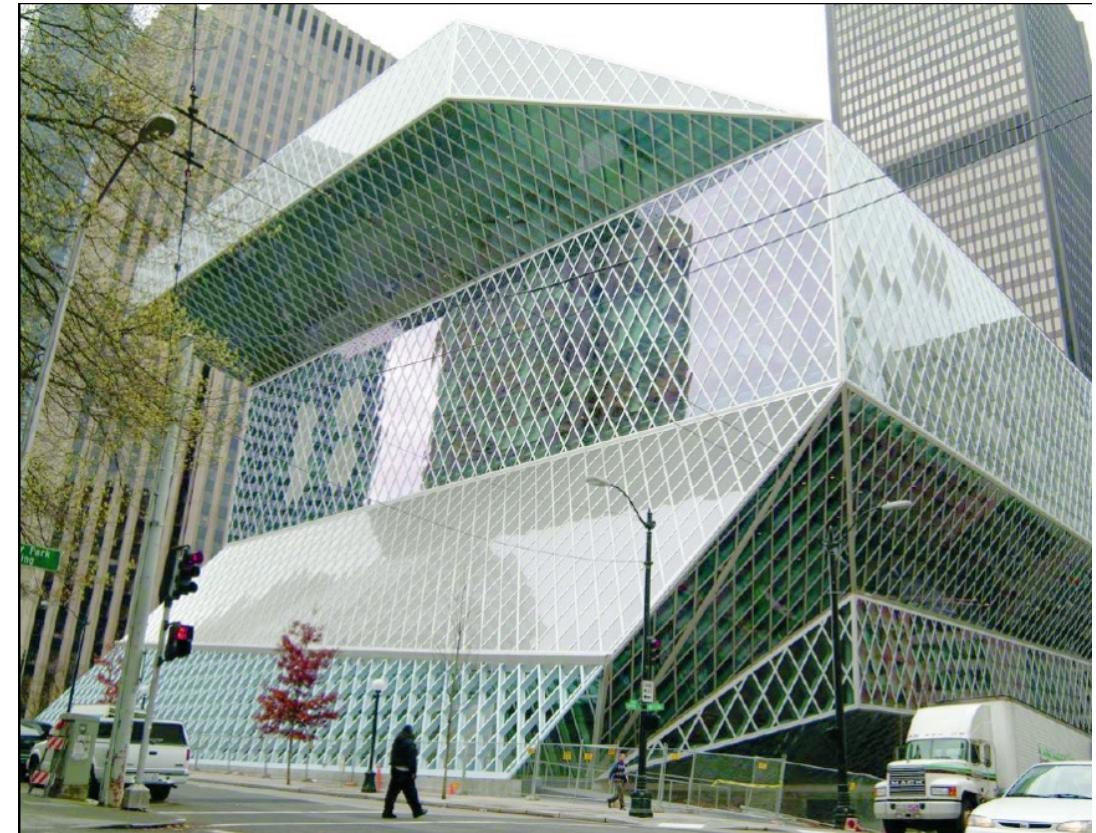
call
nov 1998



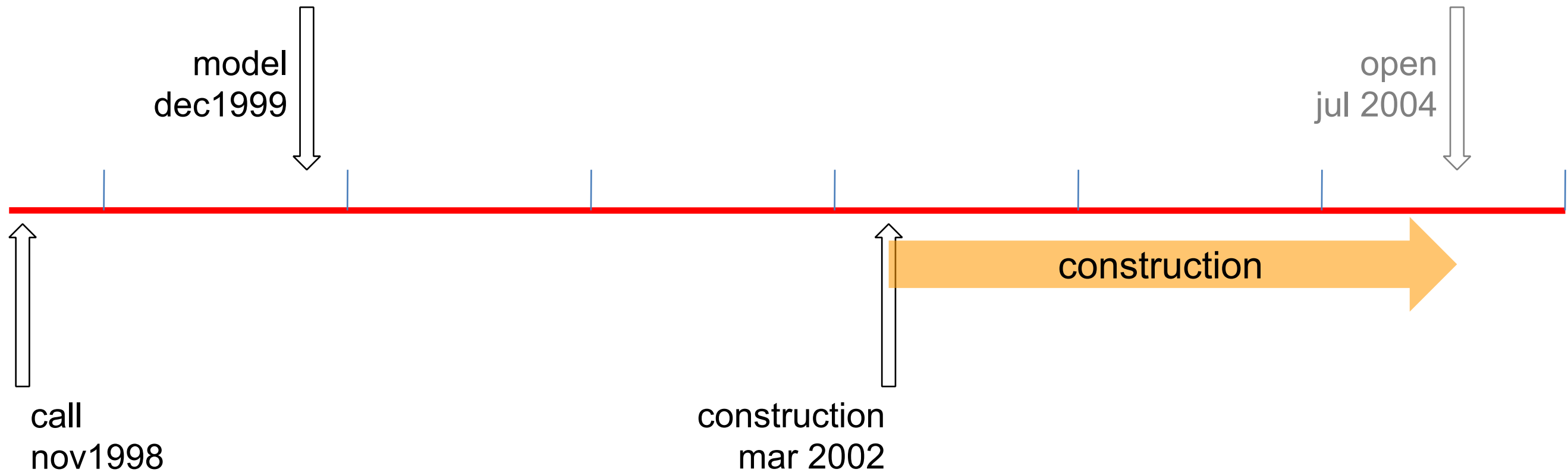
construction
mar 2002

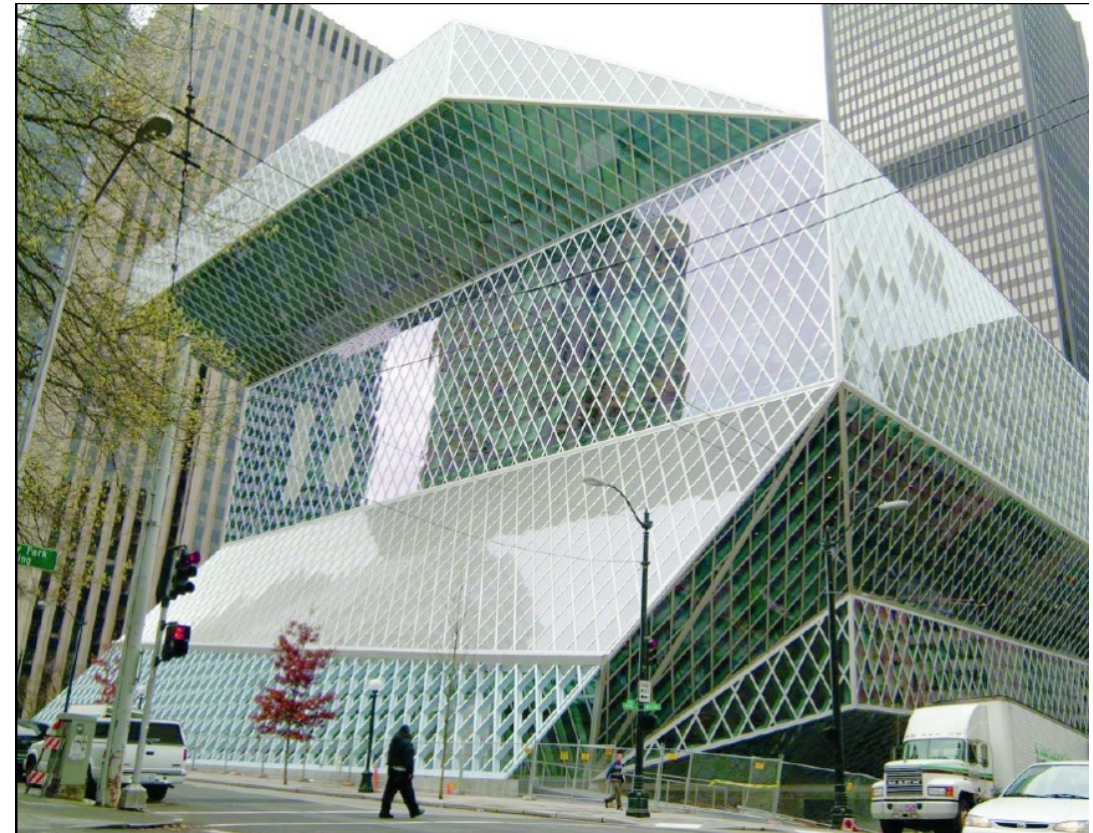
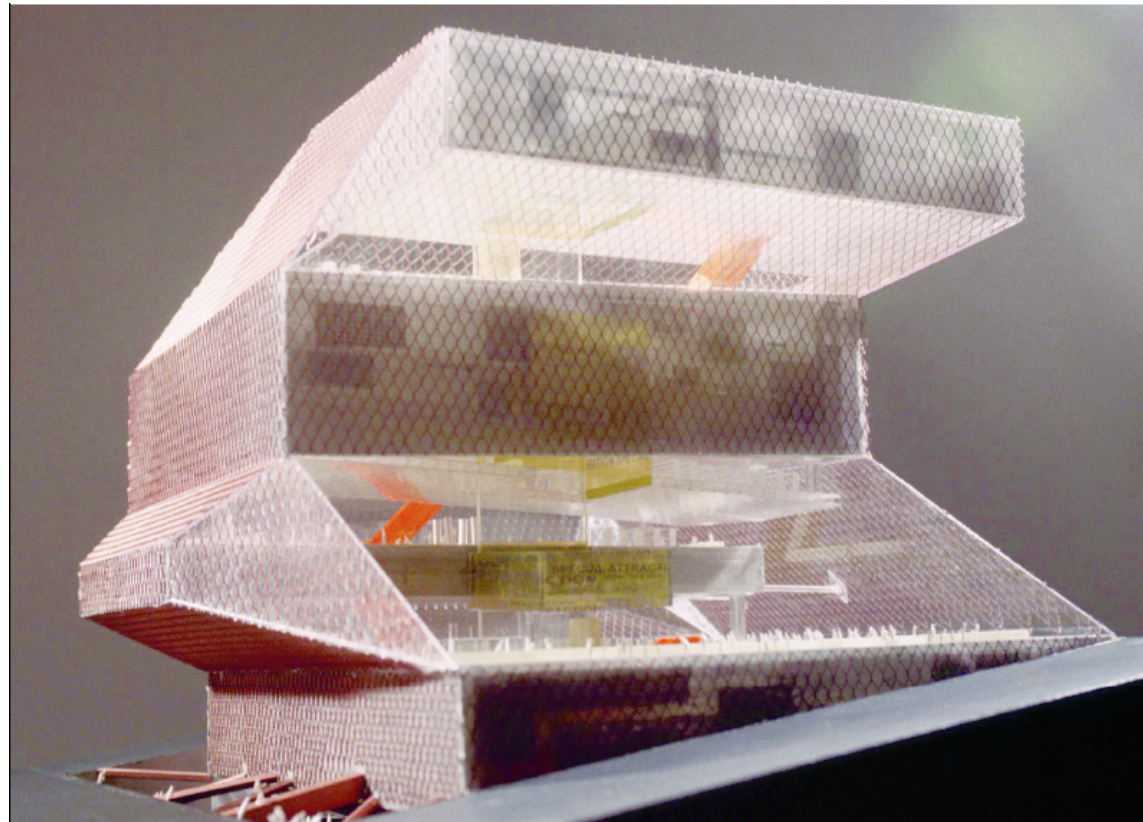


model
dec 1999



open
jul 2004





model
dec 1999

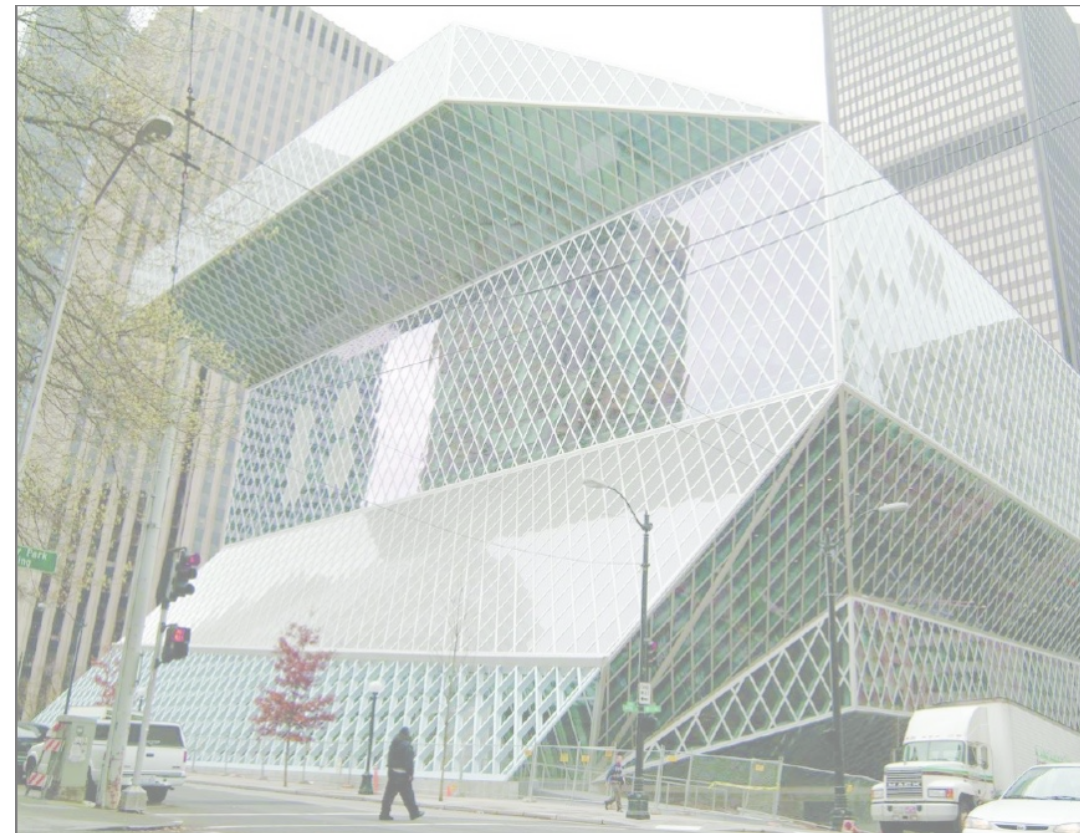
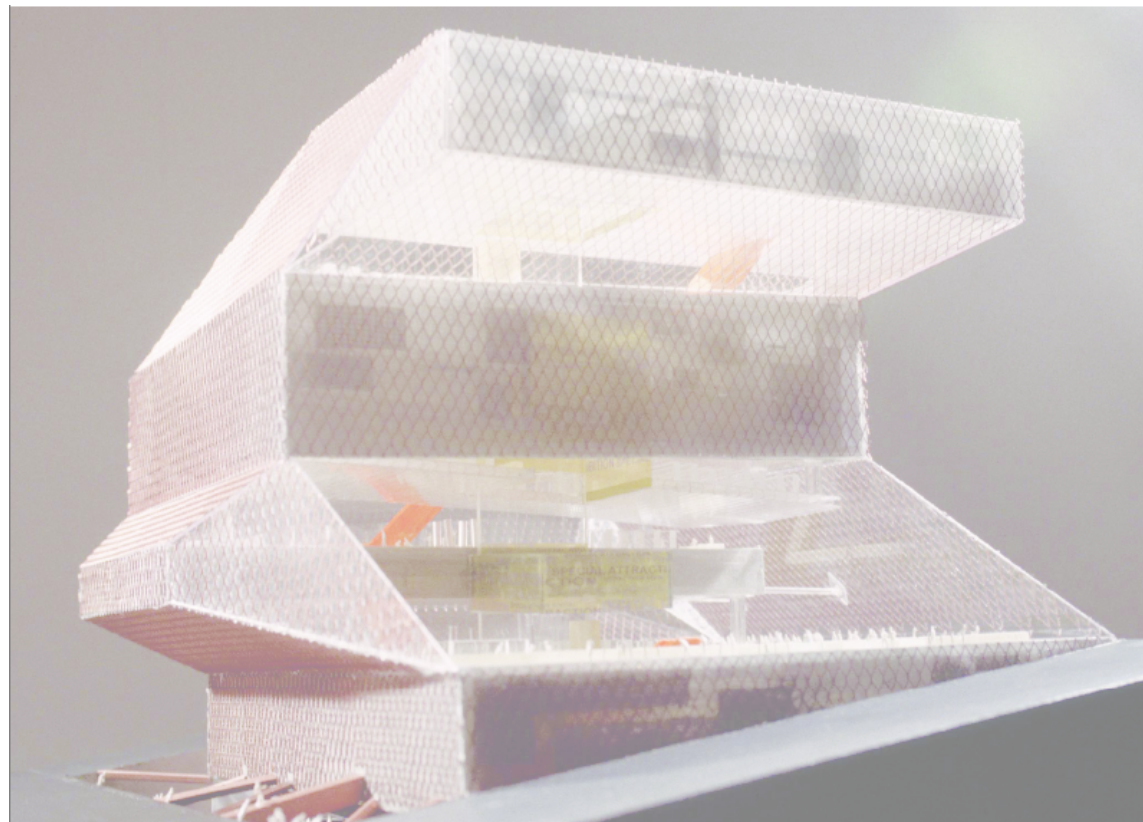
open
jul 2004

design

construction

call
nov 1998

construction
mar 2002



model
dec 1999

open
jul 2004

design

construction

call
nov 1998

construction
mar 2002

What is a prototype

Prototype

- Concrete representation of an interactive system

Goal

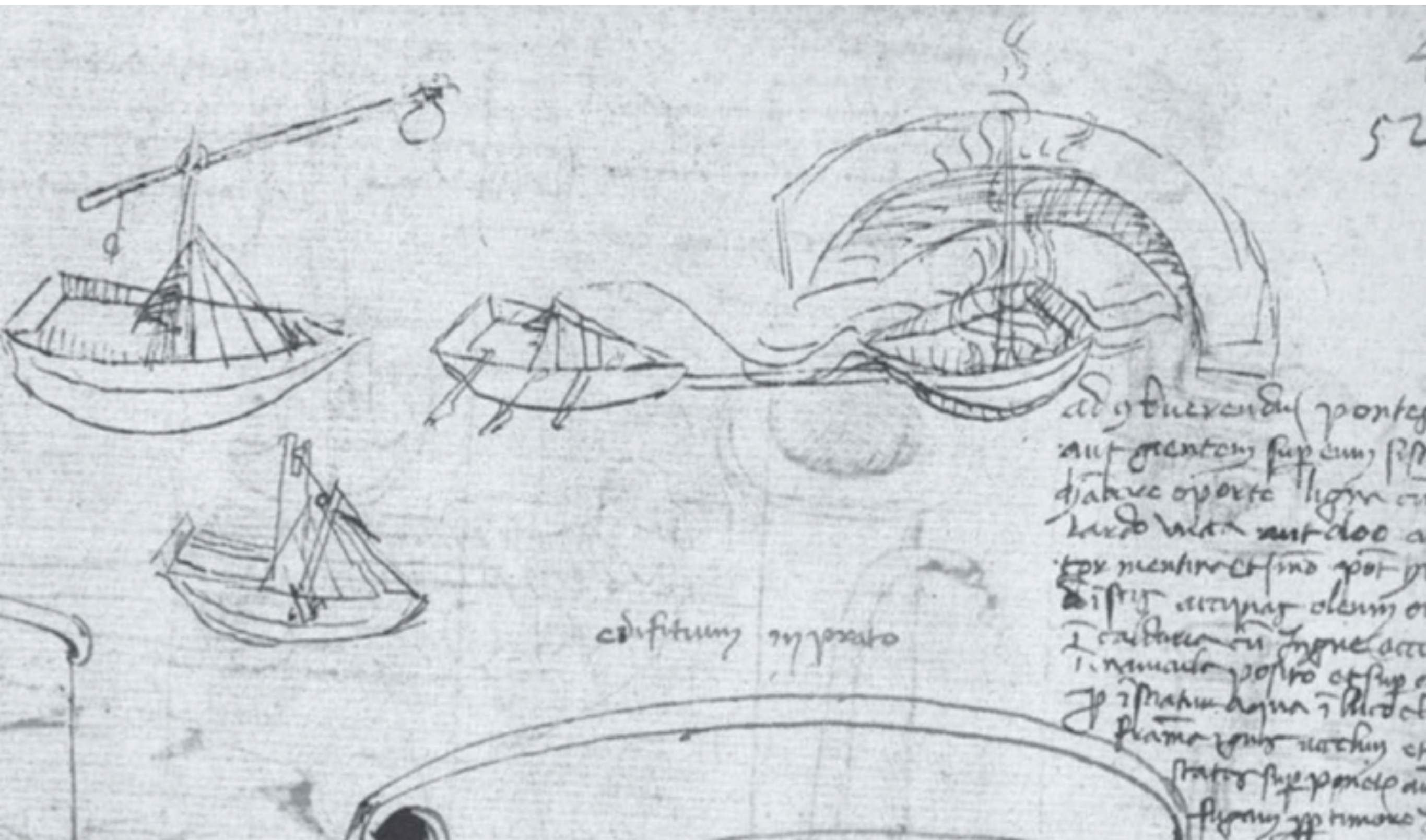
- Consider different alternatives (before implementing them)
- Ensure usability under diverse conditions
- Help users, managers, developers etc. imagine the interface
- Focus on problematic part of the interface

Choice of the prototypes

- Depends of the design phase
- Depends on designers' needs

From sketches to high-fidelity prototypes

Sketching is a tool of thought



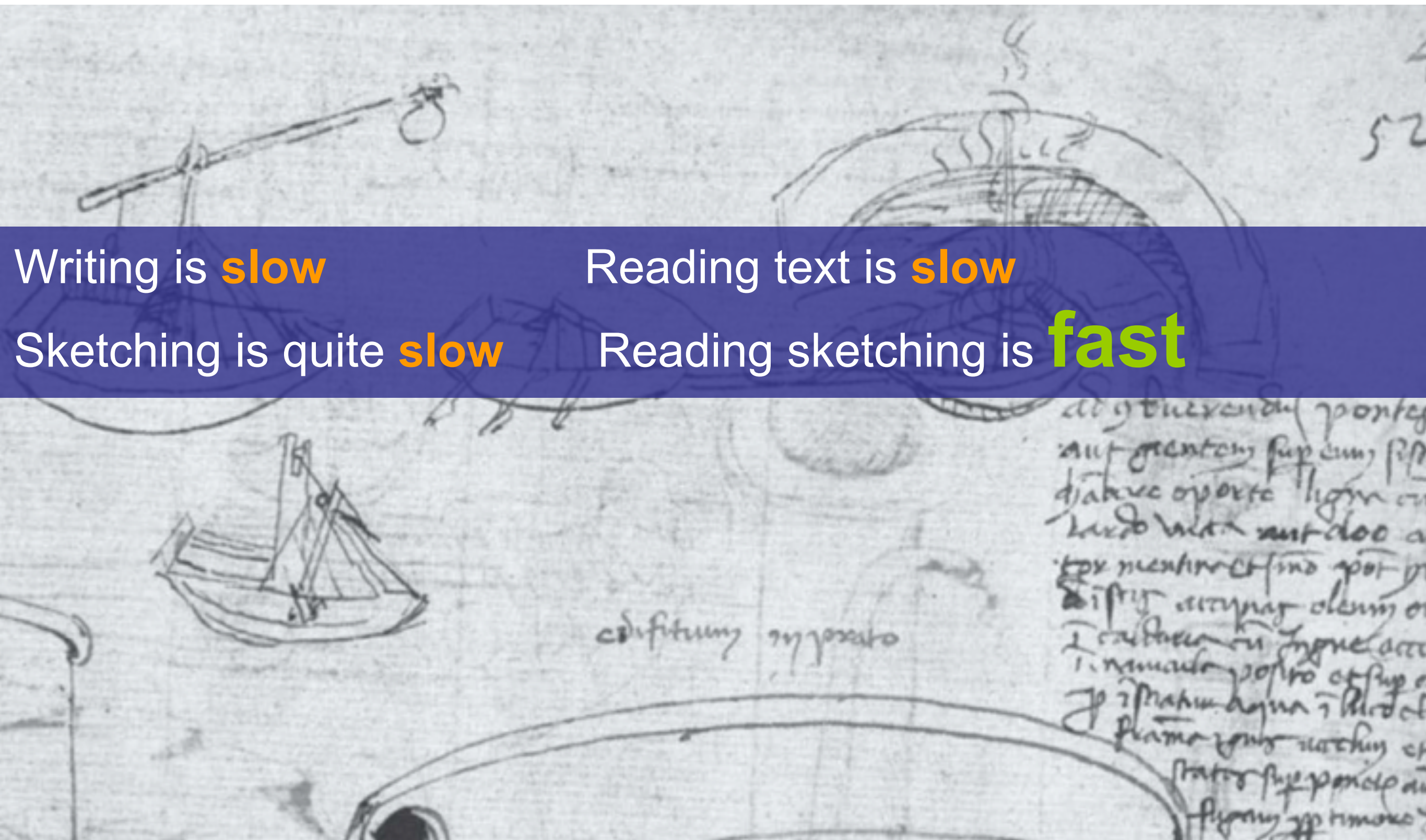
Sketching is a tool of thought

Writing is **slow**

Reading text is **slow**

Sketching is quite **slow**

Reading sketching is **fast**



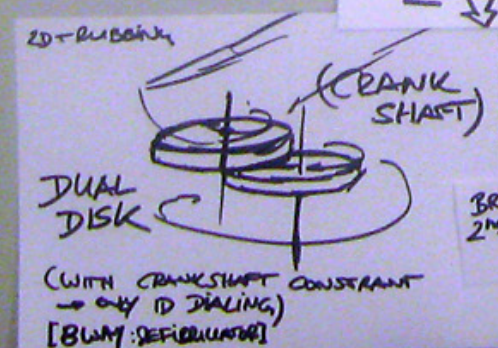
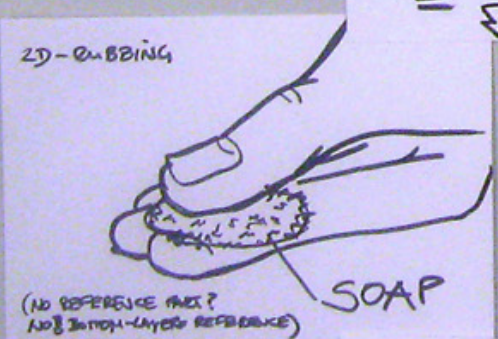
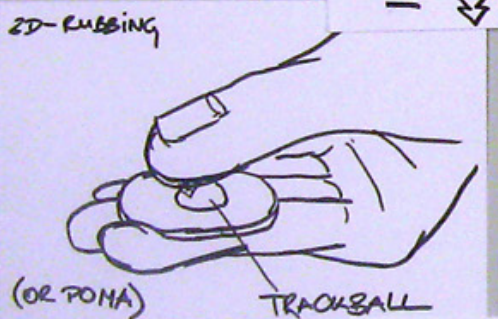
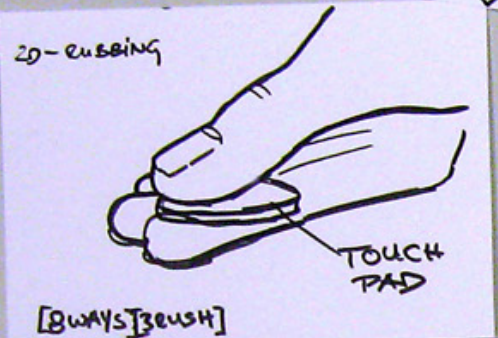
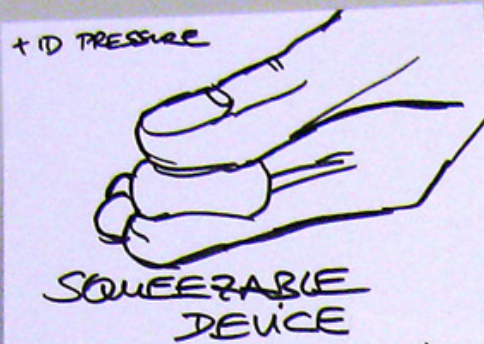
FLAT MOBILE DEVICES

TO MAKE SMALL BUT INTERACTIVE DEVICE,
WE NEED LOW VOLUME
BUT HIGH SURFACE
& LONG EDGES

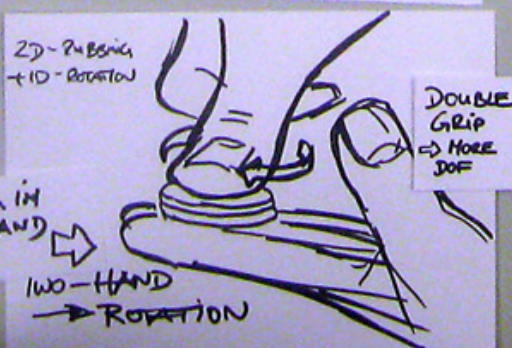


HOW TO INTERACT
WITH A FAT
DEVICE ?

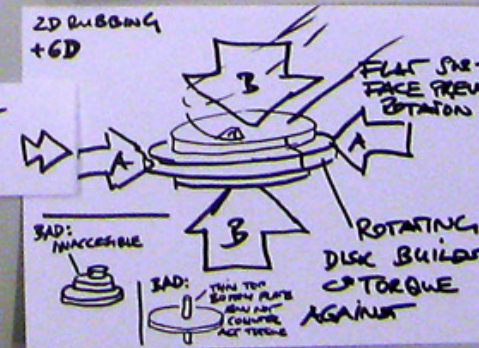
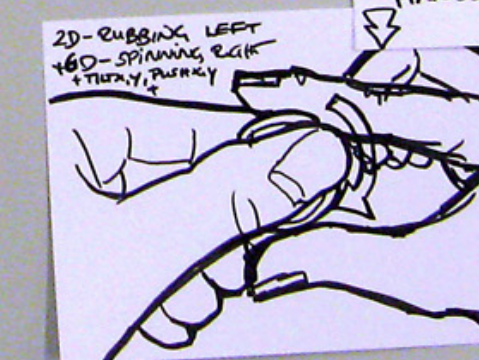
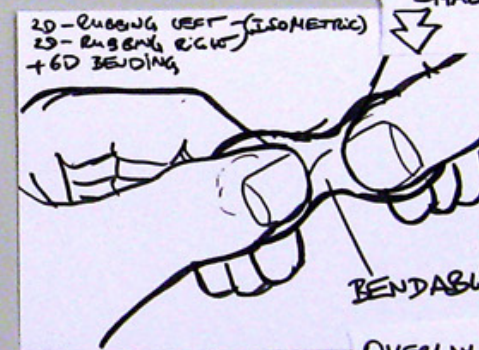
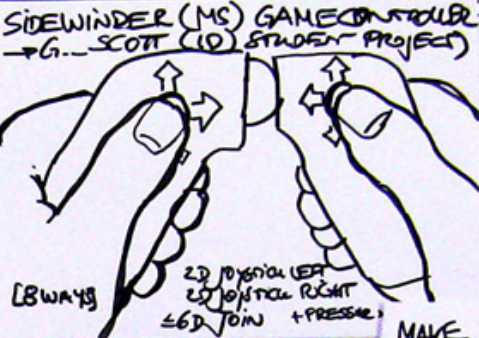
ONE-HANDED = RUBBING



TWO-HANDED

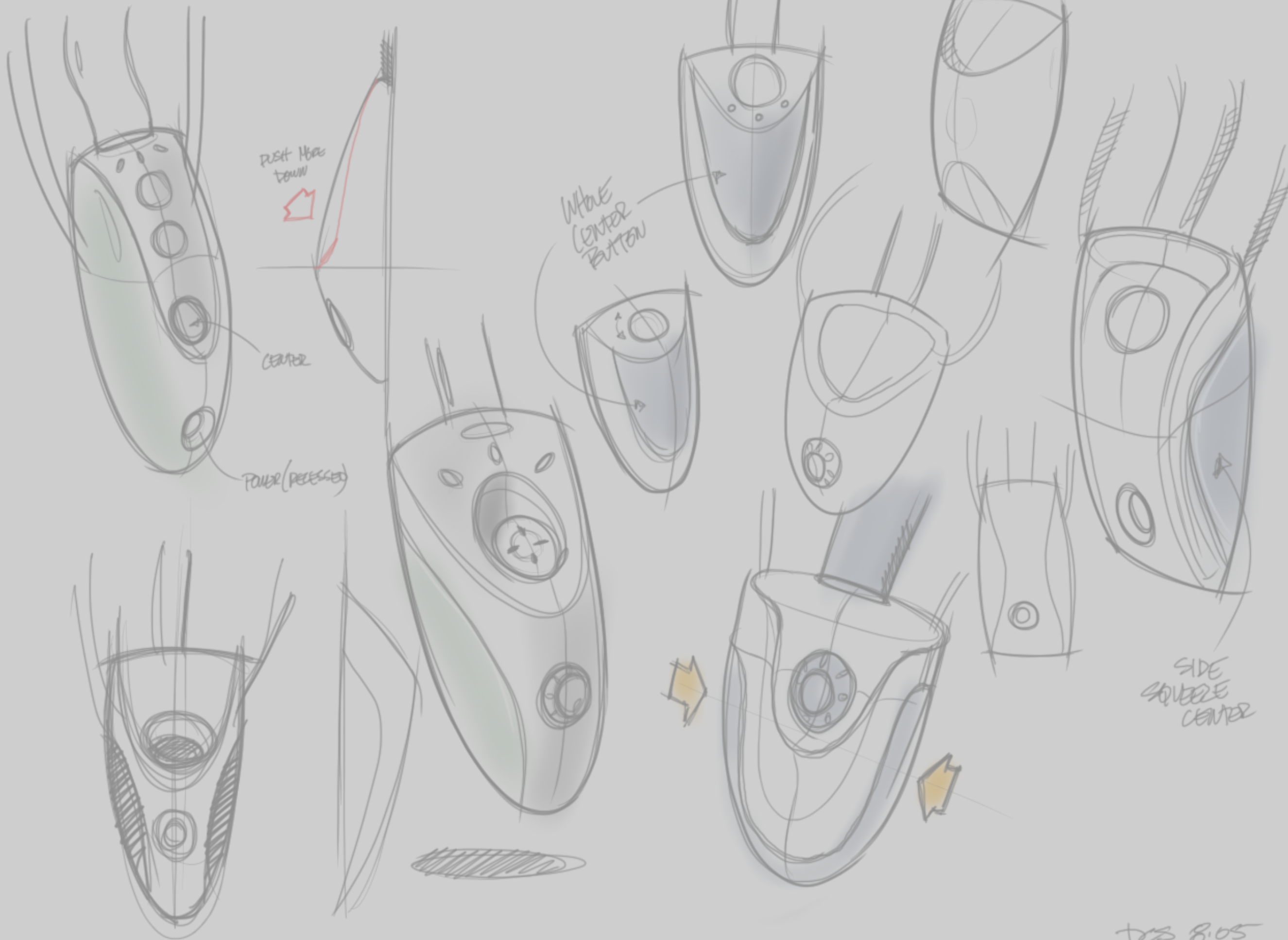


BENDING



MAKI RING





Push Here
Down



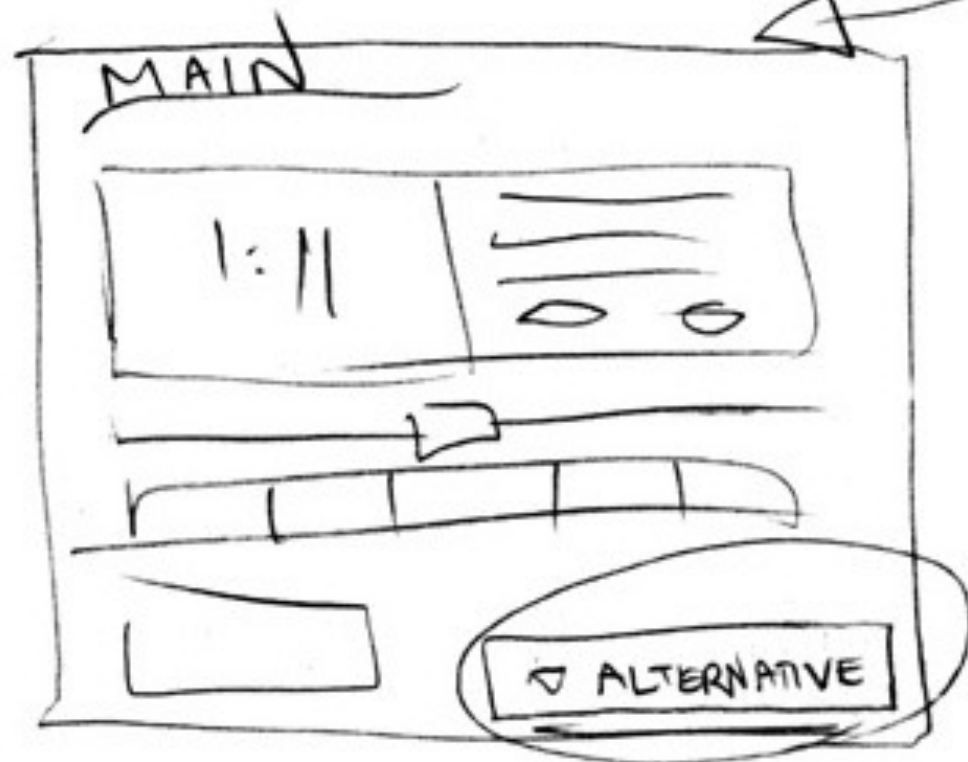
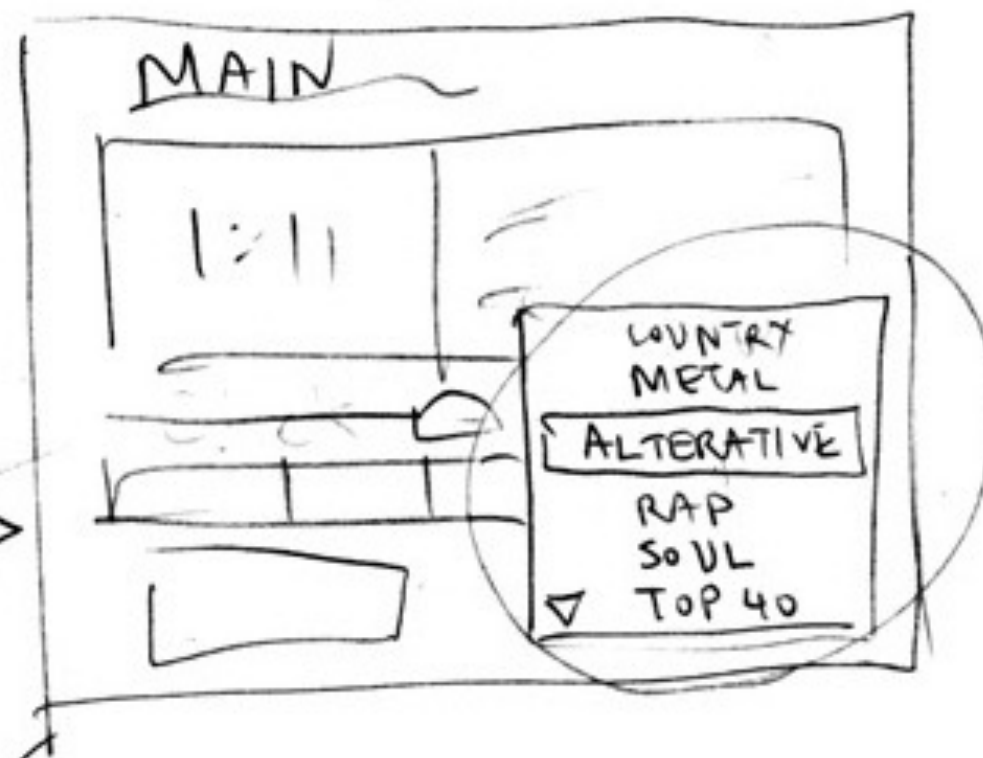
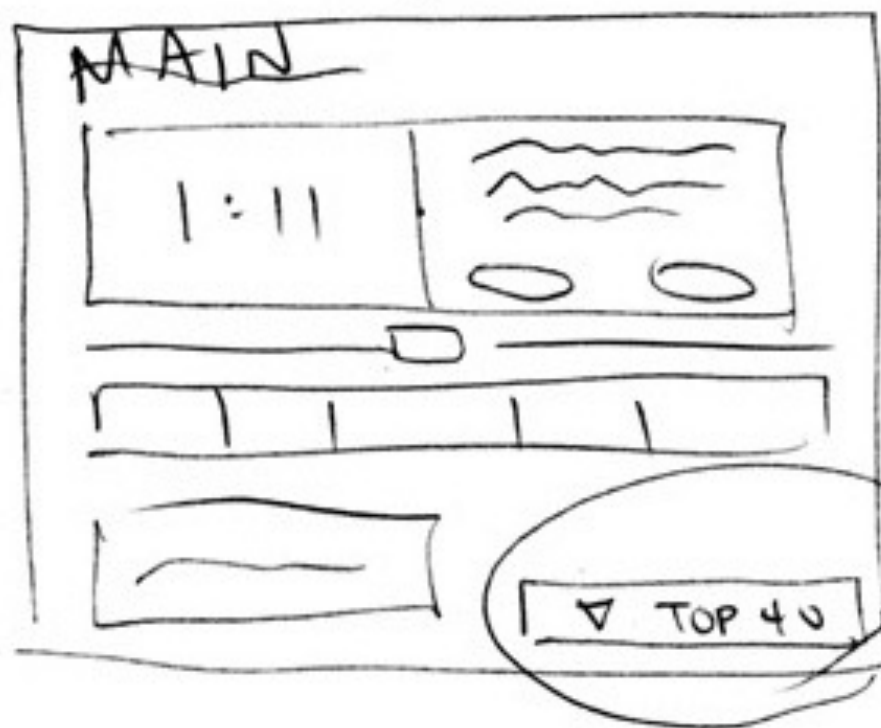
Center

Power (pressed)

White
Center
Button

SIDE
SQUEEZE
CENTER

DES 8.05



Capturez l'interaction, pas que l'interface



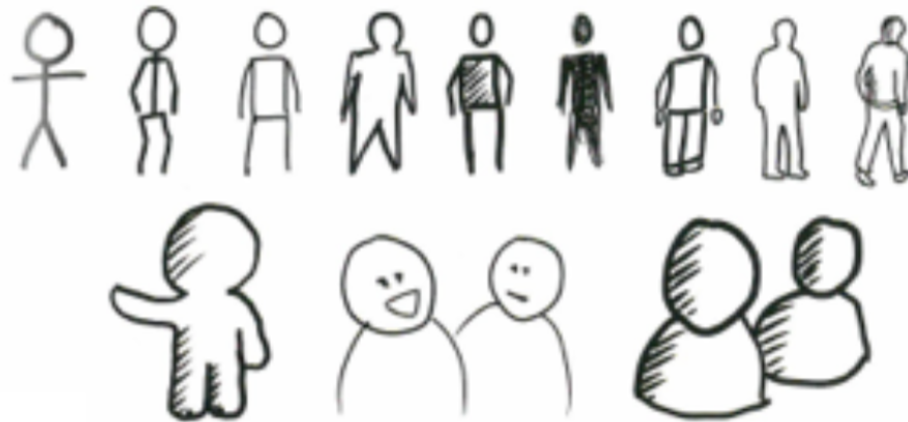
Interface



Interaction

Graphical Vocabulary

People



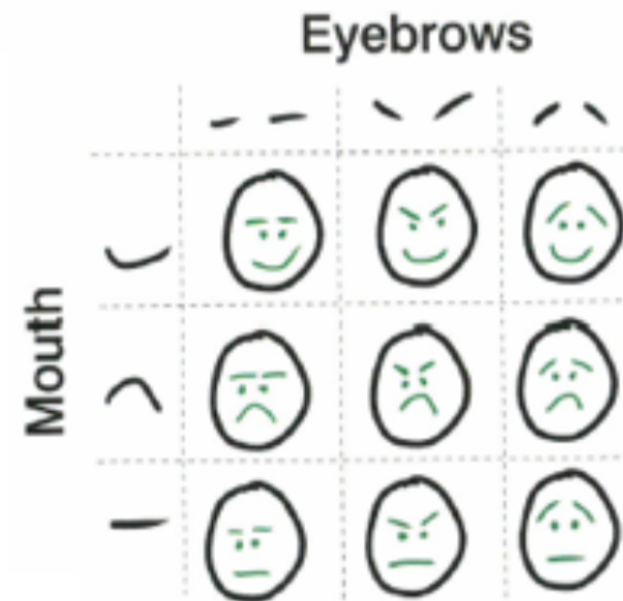
Activities



Bodies and Emotions



Faces and Emotions



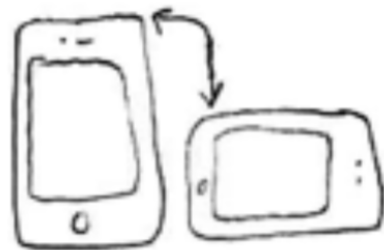
Combining Different Sketch Elements to Illustrate Situations



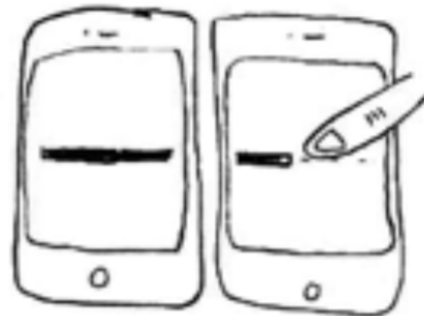
Graphical Vocabulary



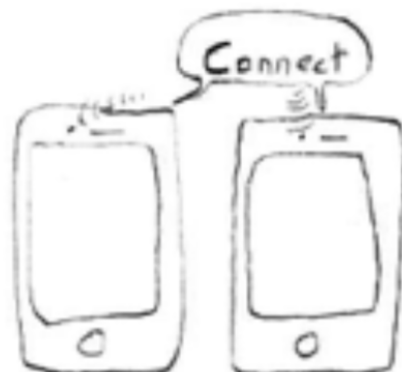
Both people type a word chosen by them



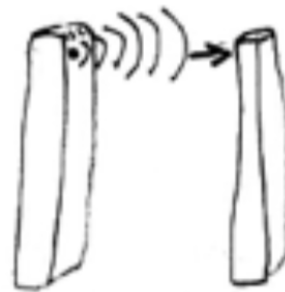
Rotate in a pattern, where other person has to mimic it (accelerometer)



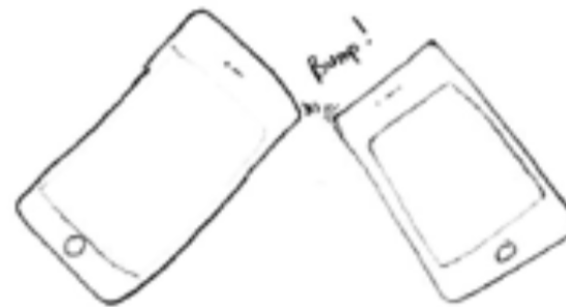
Synchronous gesture
Trace a line across both side by side devices as a single stroke



Microphones pick up spoken command at similar volume



LED strobe pattern captured by camera



Bump. Accelerometer matches bump vibrations



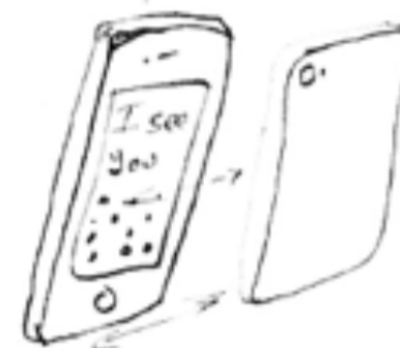
Faint musical sound played on one device picked up by the other device



Ambient light sensor
Touch surfaces together in a pattern;
Both detect same light/dark pattern



3 simultaneous taps on both phones

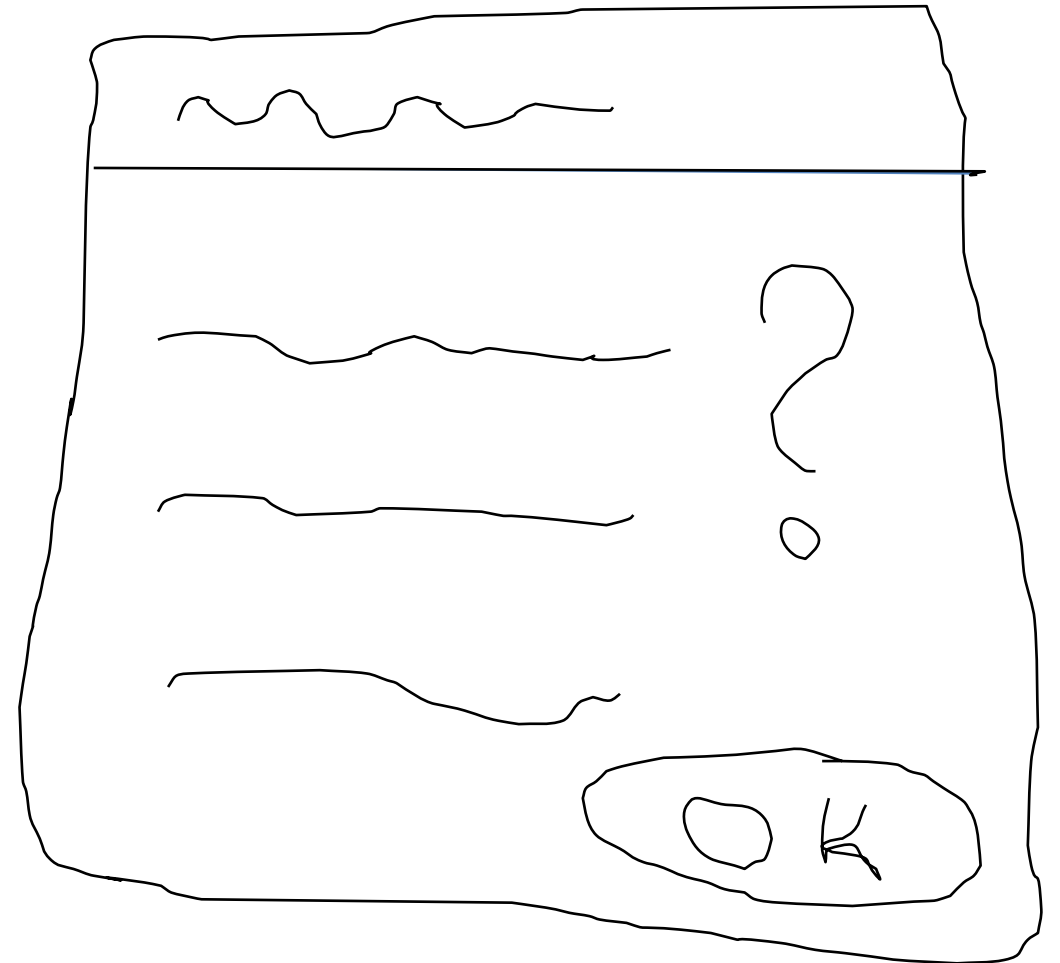


Mutual Video/photos captures identifying images such as tags in camera

The Attributes of Sketches

Quick

- To make



The Attribute of Sketches

Quick

- To make

Timely

- Provided when needed



The Attribute of Sketches

Quick

- To make

Timely

- Provided when needed

Disposable

- Investment in the concept, not the execution
- Inexpensive



The Attribute of Sketches

Quick

- To make

Timely

- Provided when needed

Disposable

- Investment in the concept,
not the execution

- Inexpensive

Plentiful

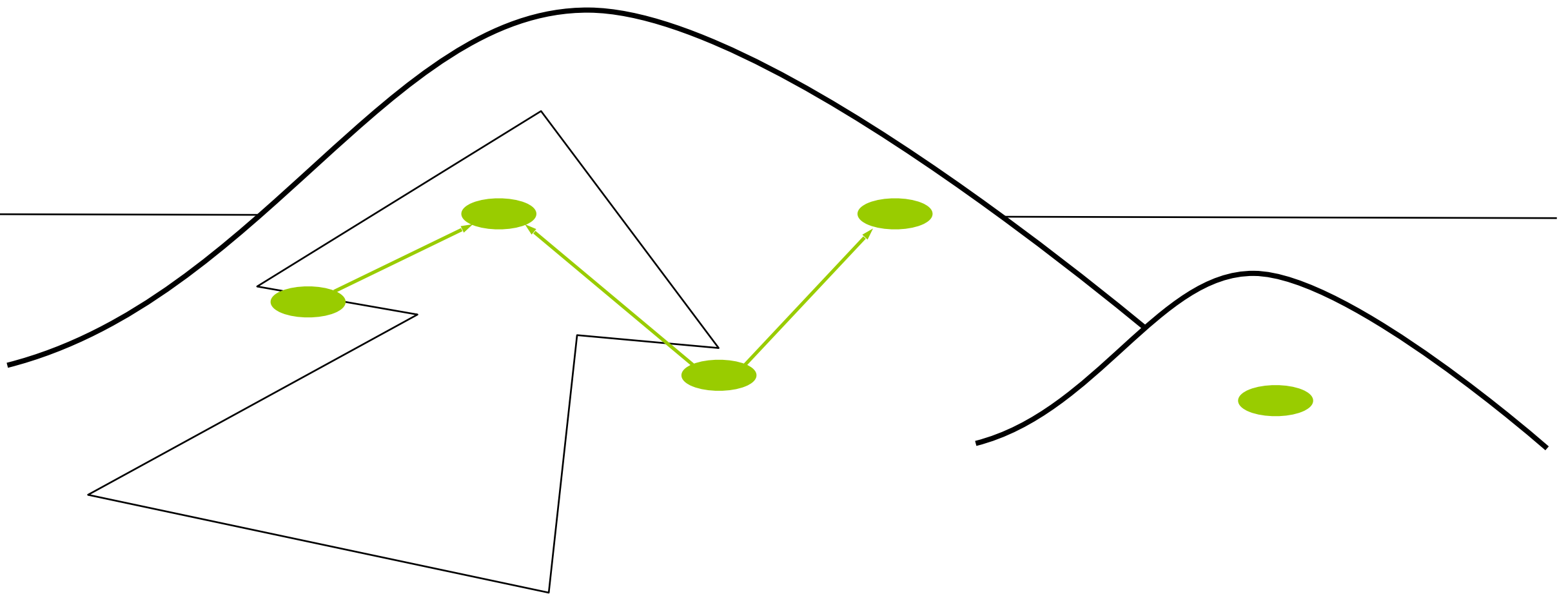
- They make sense in a
a collection or series
of ideas.



Design Process

- (1) create k new designs;
- (2) drop $k-1$ worst designs

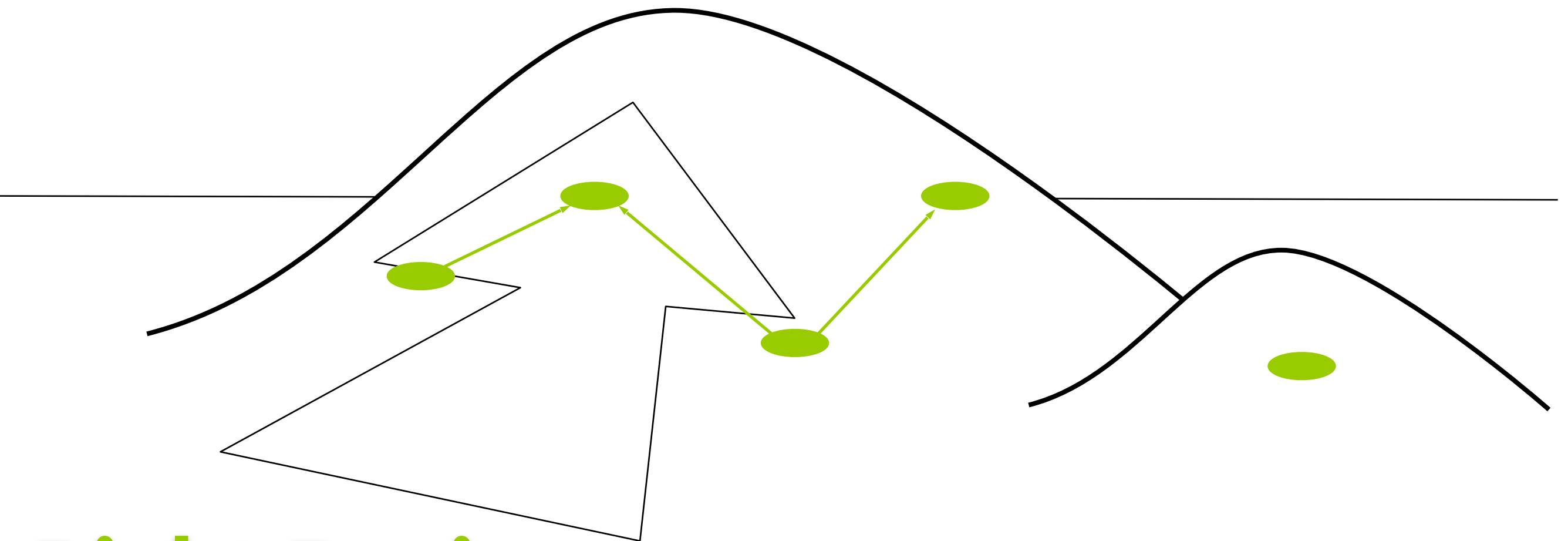
this process finds the tops of multiple hills
and works with “distracter” hills



Design Process

- (1) create k new designs;
- (2) drop $k-1$ worst designs

this process finds the **tops** of multiple hills
and works with “distracter” hills

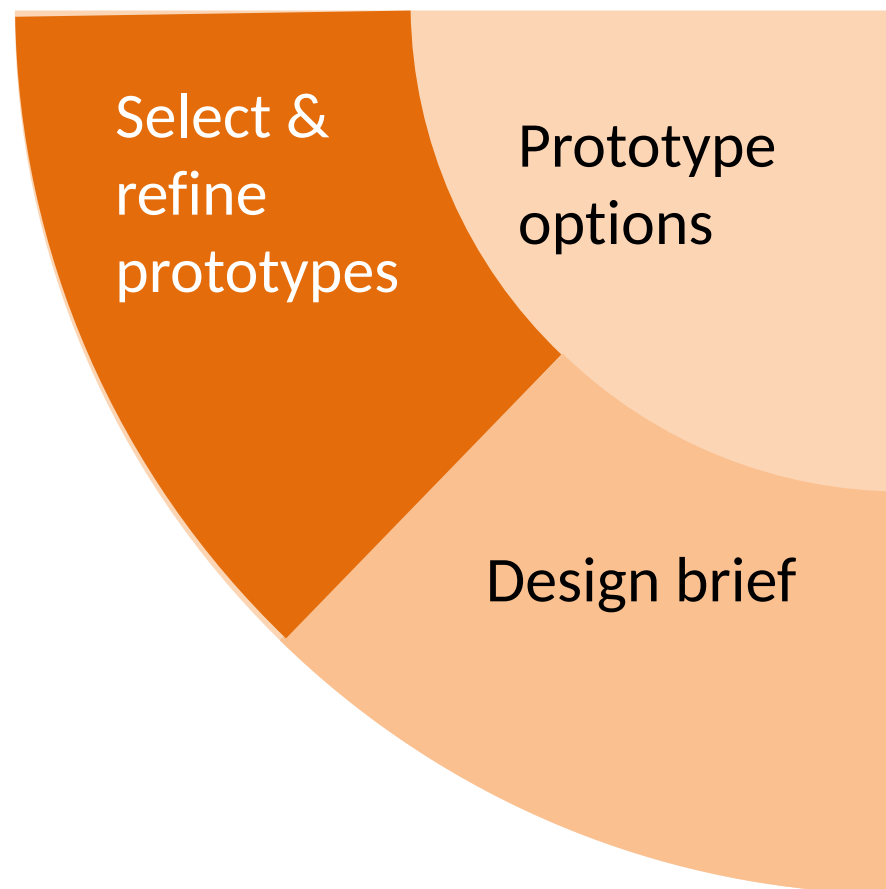


Right Design

Design Resources

- Low-fidelity prototype
- Medium-fidelity prototype
- High-fidelity prototype

Analyze information



Collect information

- sketches

What should it be?

Design

Starting point



Ideas generate more ideas

Focal point



Ideas selected and refined

Sketch



Prototype

Low investment
More opportunities to explore
Fail early... learn

What is a prototype

Prototype

- More Concrete representation of an interactive system
- Focus more on quality than quantity

Goal

- Consider different alternatives (before implementing them)
- Ensure usability under diverse conditions
- Help users, managers, developers etc. imagine the interface
- Focus on problematic part of the interface

Prototypes

Level of details

Level of interactivity

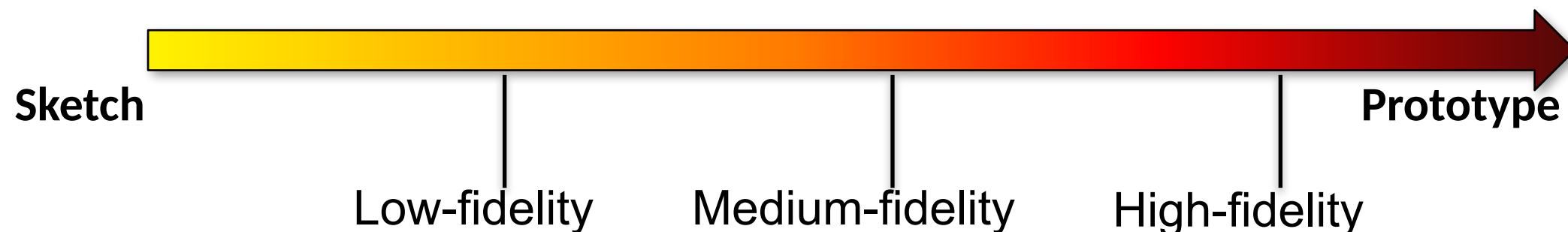
Prototyping strategies

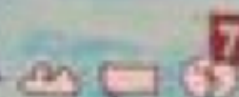
Levels of details

Low-fidelity prototyping

Medium-fidelity prototyping

High-fidelity prototyping





Edit my Profile

News Feed



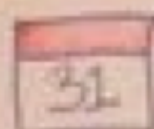
Messages

25



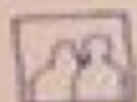
Friends

5



Events

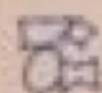
1



Business no hours



Create a Group
see all



Application

3

More v

1

News Feed

Top News [Add Event](#)

300+

What's on your mind?



CARDEIRA BEGRO
PATRADO TOMORROW!



SABA WZIA AWON POR AGUI
COM SEU QUALIFICO
COMENDO CAPIM.



IT WAS A LOOOONG DAY...!



BRAH.
DON'T TRY THIS AT HOME



SWEET LIKE CANDY TO MY
SOUL / SWEET YOU ROCK /
AND SWEET YOU ROLL.



oh yeah I'm AWESOME!

TEXTS

What's on your mind?



Facebook



Facebook

Facebook

Facebook

Go to the home page

Facebook

Facebook

Facebook

Facebook

Facebook

Facebook

Facebook

Facebook

Facebook

Verizon	(212) 890.	Contacts	Em
No Categories			
#a	b	cde	fghijk
lmn	opq	rst	uvwxyz
Baldeschwieler dl@NAVIMATION... e			
DSL, Verizon (877) 483-1794 w			
JPMorgan Cha...			
Ken Scott		Kscott@randrasone	
Unnamed			
Verizon		(212) 890-6611 w	
New View Tools			

Thumbnails
Create Copy
Delete Contact
Send E-mail to Contact
Beam Contact...
New View Tools

Contacts ≡ m ⊗

No Categories ▾

#	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
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Bakke Schwieler dl@NAVIMATION... e
 DS' ... (877)483-1794. w

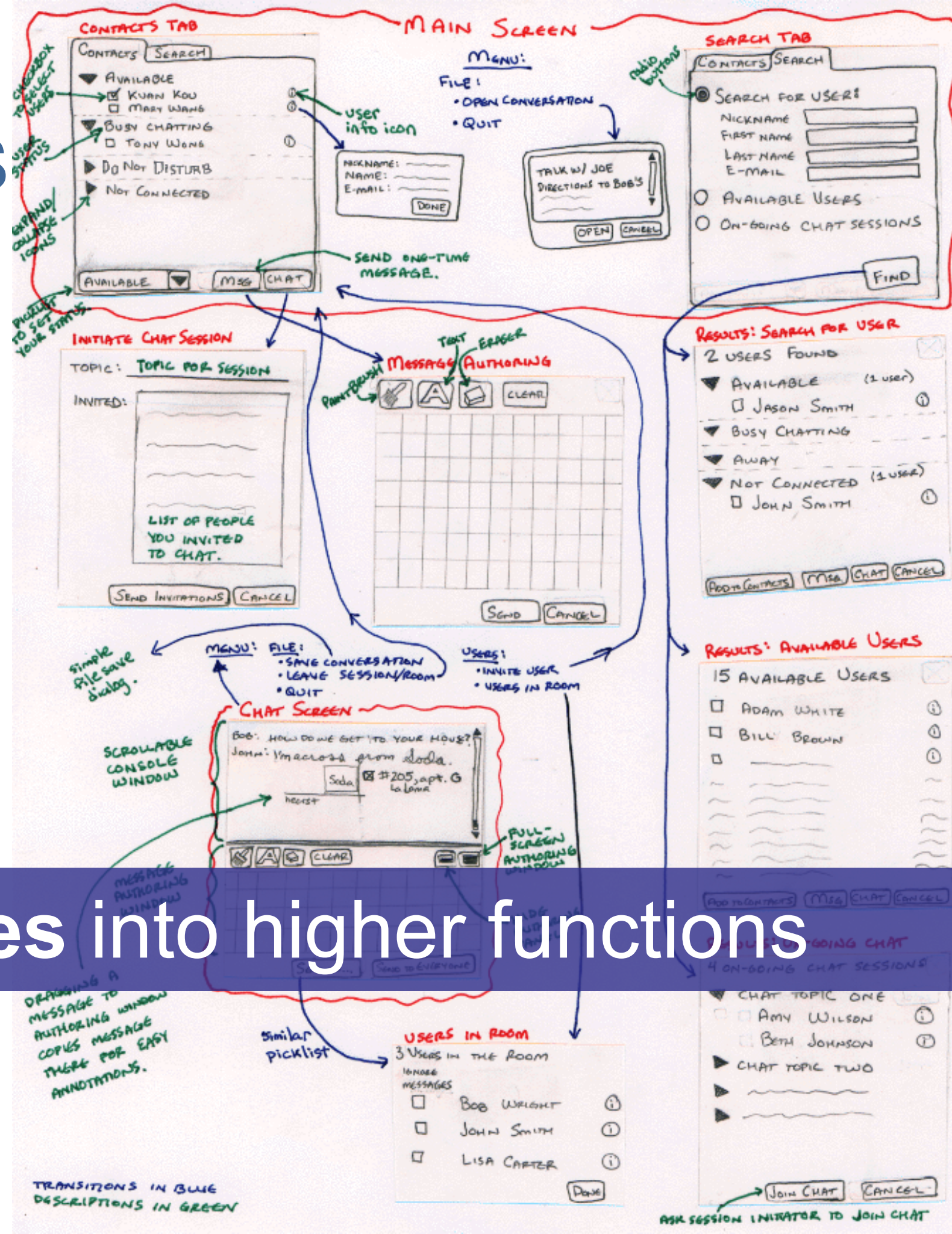
JP. Contacts

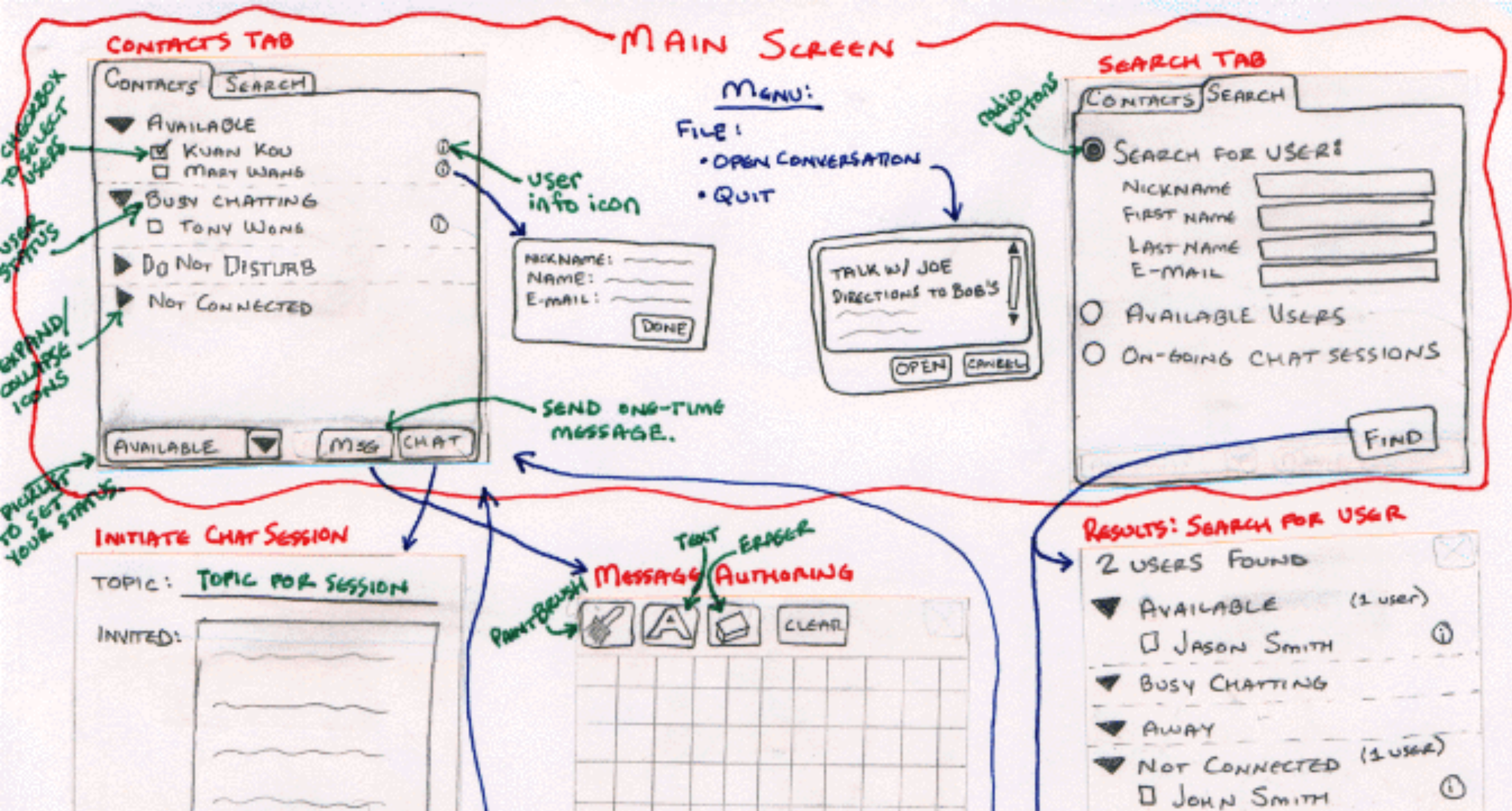
Ke ⚠ Contacts will be permanently deleted.
 Continue?

Yes
No

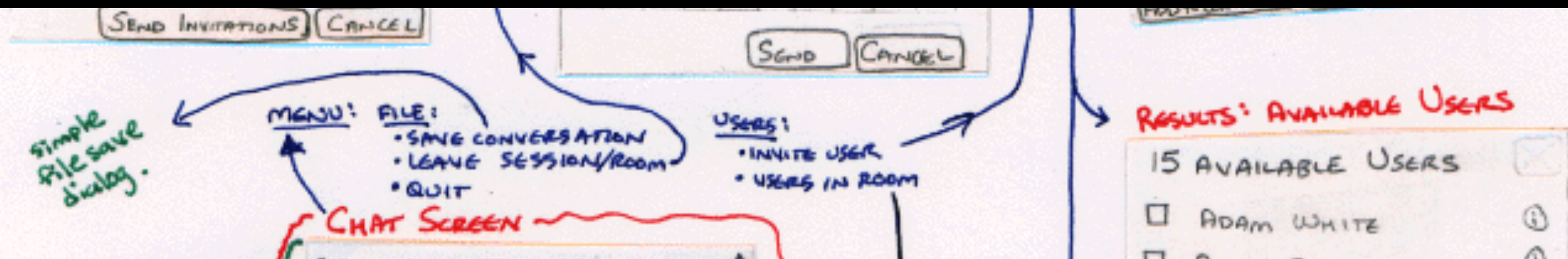
New View Tools 🔍 ▴

Link diagrams





here: **blue = links** between pages

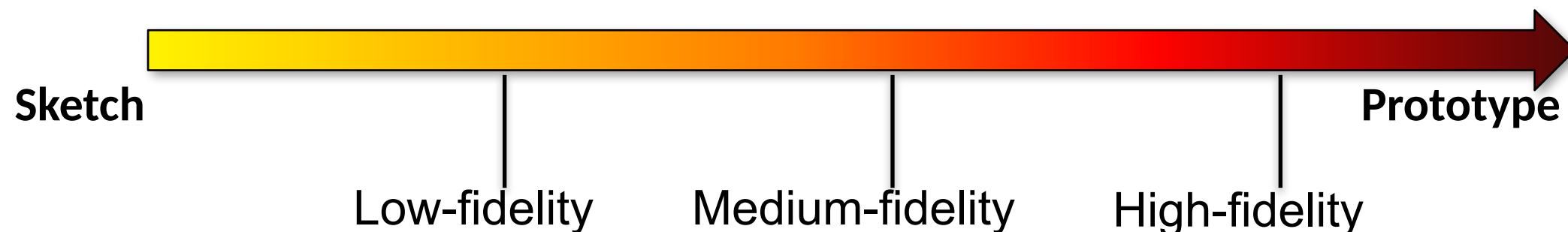


Levels of details

Low-fidelity prototyping

Medium-fidelity prototyping

High-fidelity prototyping



On-line prototypes

Use the computer, longer to create, more polished

More appropriate later in the design process

Examples

- Animations
- Interactive videos
- Scripting languages
- Interface builders

Fidélité	Avantages	Désavantages
Basse	- Coût faible - Évaluer plusieurs designs - Outil de communication - Remaniable - Utile pour identifier le marché - Preuve de concept	- Vérification d'erreurs difficile - Spécifications peu détaillées - Utilité limitée qd les spécifs sont choisies - Utilité limitée pour les tests d'utilisabilité - Navigation et flot limitées
Haute	- Fonctionnel - Interactif - Orienté utilisateur - Schéma de navigation clair - Utile pour tests - Look & feel « final » - Spécifications vivantes - Outil de vente et marketing	- Cher - Long à fabriquer - Pas efficace pour preuve de concept - Pas efficace pour choisir les spécifs

Good prototypes

Allow users to envision the final system...
... but also feel comfortable suggesting changes

Prototypes

Level of details

Level of interactivity

Prototyping strategies

Level of interaction

Non interactive

- Fixed
- Can still show interaction...
- Examples: Images, videos

Low interaction

- Pre-determined path
- Can test alternative forms of interaction
- Example: Sequence of screenshots

High interaction

- Open
- Users interact with the system, with some limitations
- Example: Wizard-of-Oz

Contacts									
No Categories ▾									
#	a	b	c	d	e	f	g	h	i
Bal	d	e	s	c	h	w	i	e	r
d	e	s	c	h	w	i	e	r	
d	e	s	c	h	w	i	e	r	
d	e	s	c	h	w	i	e	r	
d	e	s	c	h	w	i	e	r	
d	e	s	c	h	w	i	e	r	
d	e	s	c	h	w	i	e	r	
d	e	s	c	h	w	i	e	r	
d	e	s	c	h	w	i	e	r	

Verizon (212) 890-6611 w

Contacts									
No Categories ▾									
#	a	b	c	d	e	f	g	h	i
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d	e	s	c	h	w	i	e	r	
d	e	s	c	h	w	i	e	r	
d	e	s	c	h	w	i	e	r	

New View Tools

Contacts									
No Categories ▾									
#	a	b	c	d	e	f	g	h	i
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d	e	s	c	h	w	i	e	r	
d	e	s	c	h	w	i	e	r	
d	e	s	c	h	w	i	e	r	

Unnamed									
Create Copy									
Delete Contact									
Send E-mail to Con									
Beam Contact...									

New View Tools

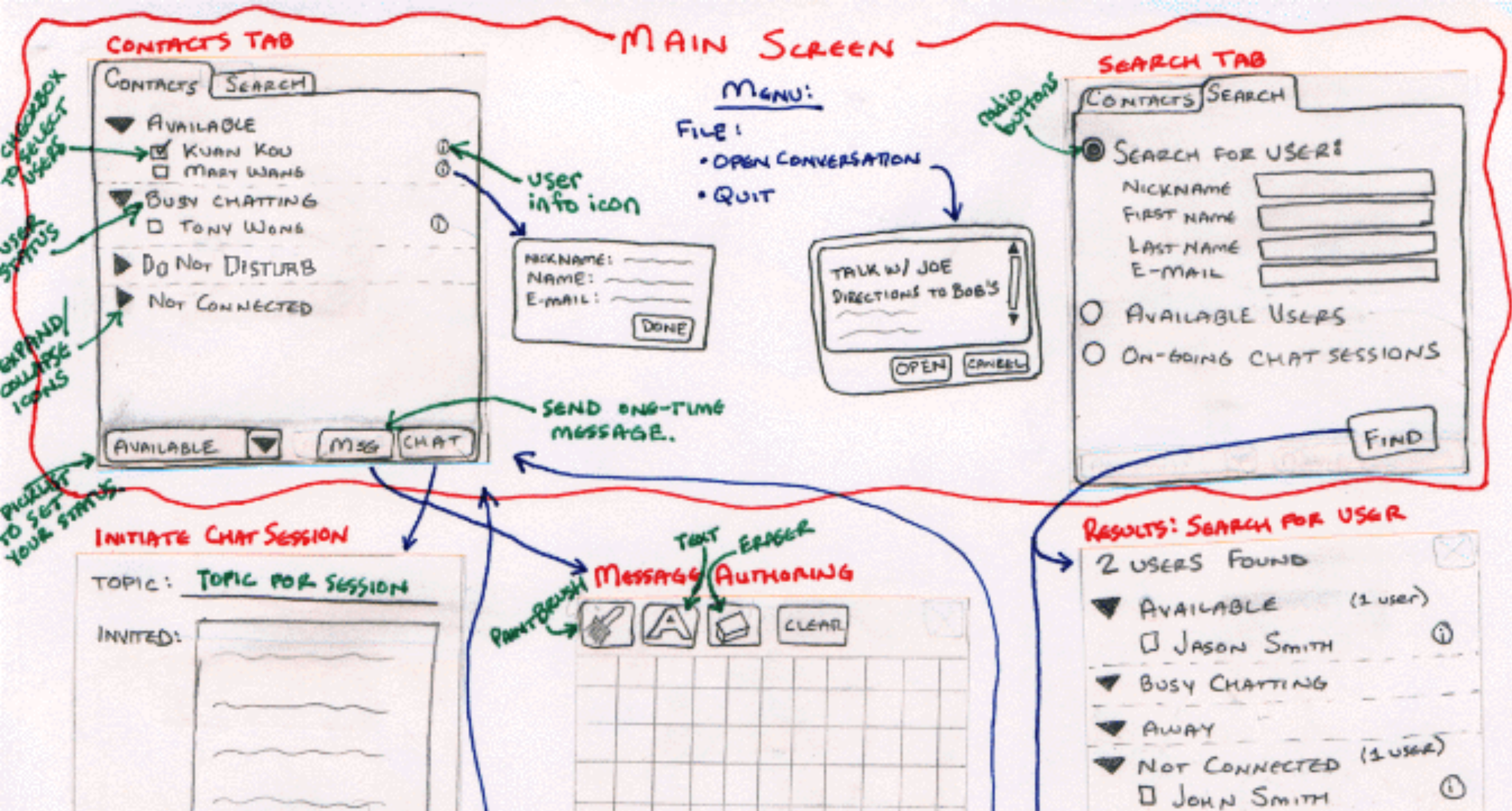
Contacts									
No Categories ▾									
#	a	b	c	d	e	f	g	h	i
Bal	d	e	s	c	h	w	i	e	r
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d	e	s	c	h	w	i	e	r	
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d	e	s	c	h	w	i	e	r	
d	e	s	c	h	w	i	e	r	

Contacts									
⚠ Contacts will be permanently deleted. Continue?									
Yes No									

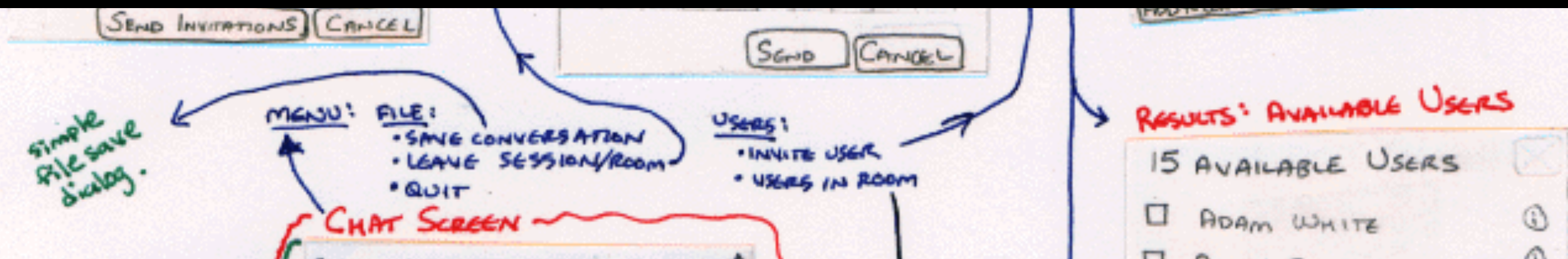
New View Tools

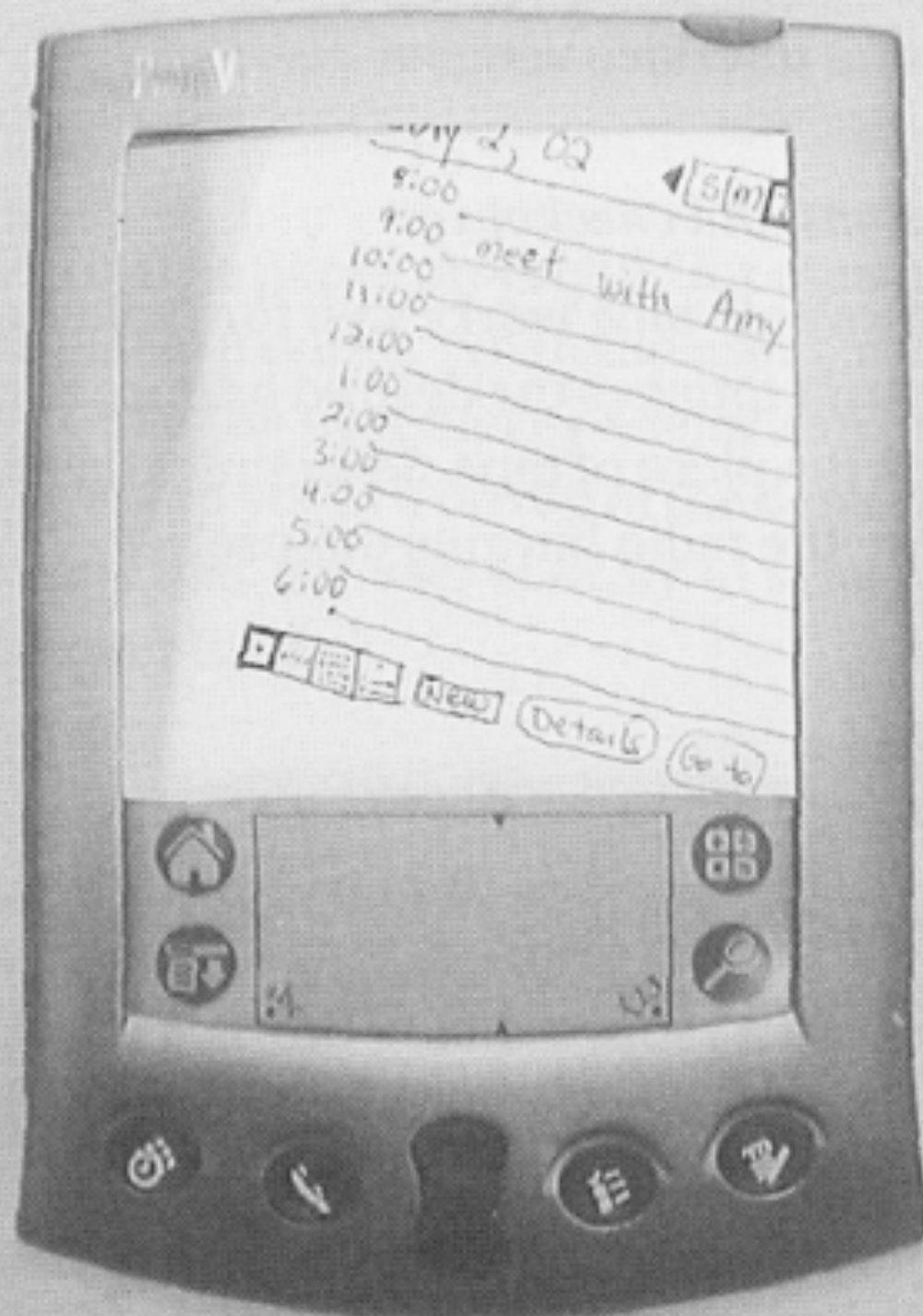
New View Tools

Fixed

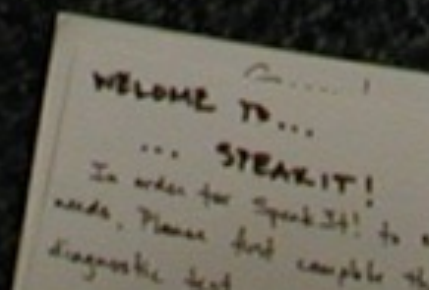
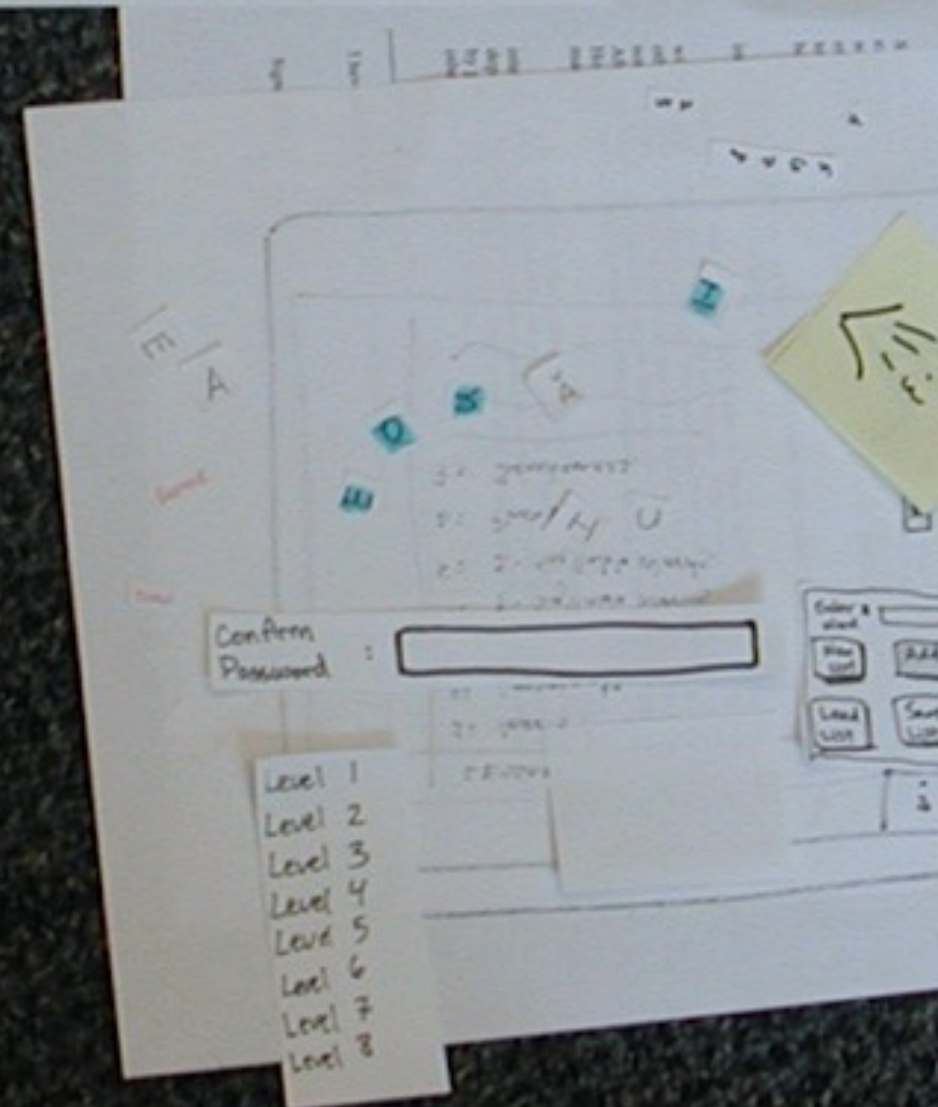
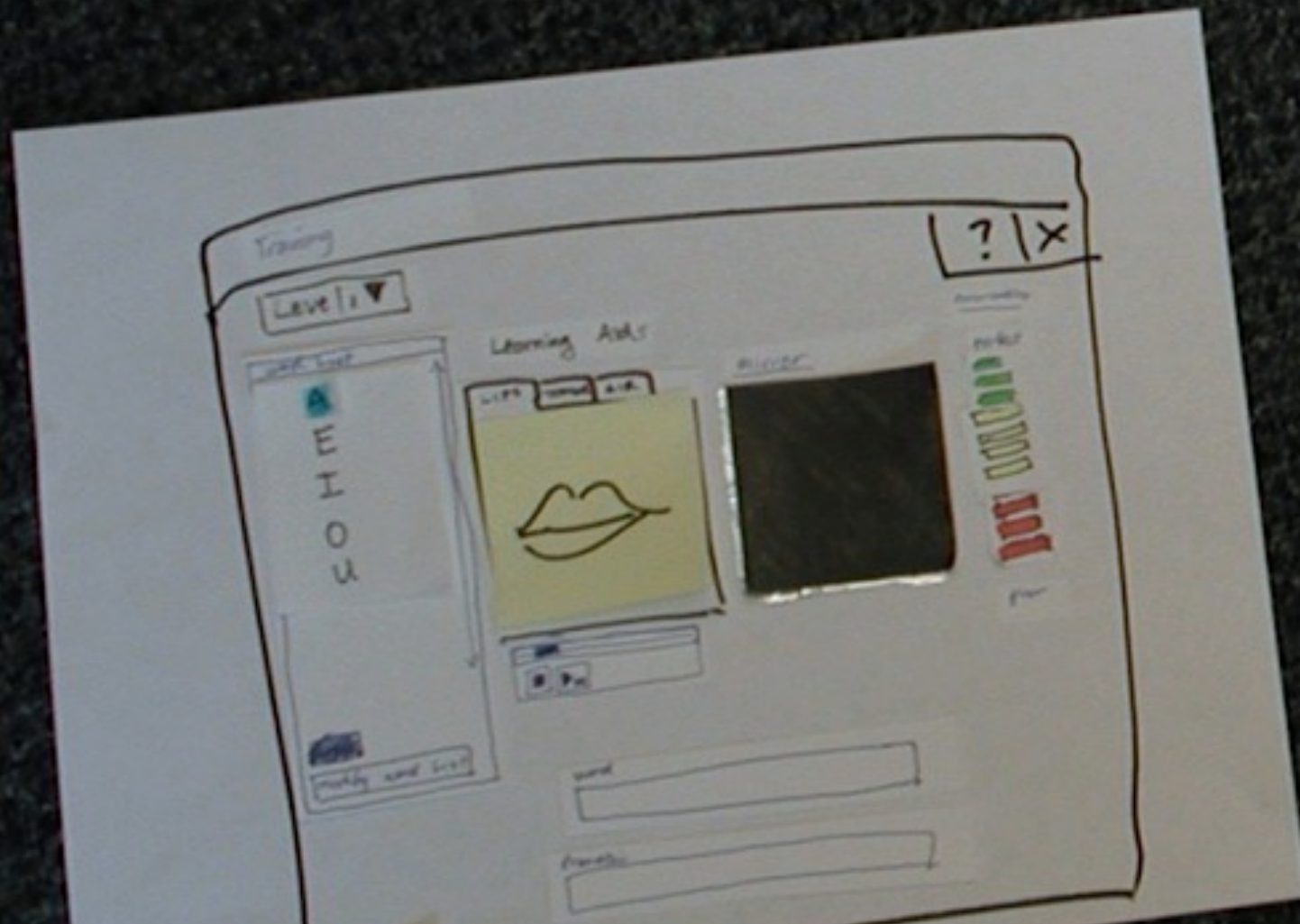


Fixed





Low interaction: add post-its etc for interactive widgets



Arrange them so as to have them **ready**



interviewer

**computer
(Wizard of Oz)**

participant

Play it out with paper as if it was real

Wizard of Oz

Technique for prototyping novel user interfaces

Designer 'plays computer'

- To create an interactive experience for the user

Useful for creating

- video prototypes
- Live experiences that rapidly explore different design alternatives

An example where this technique is very powerful?

Prototyping strategies

Vertical prototypes

- Includes in-depth functionality for a single feature
- Ex : spelling checker

Horizontal prototypes

- The entire surface interface with no underlying functionality
- A simulation; no real work can be performed

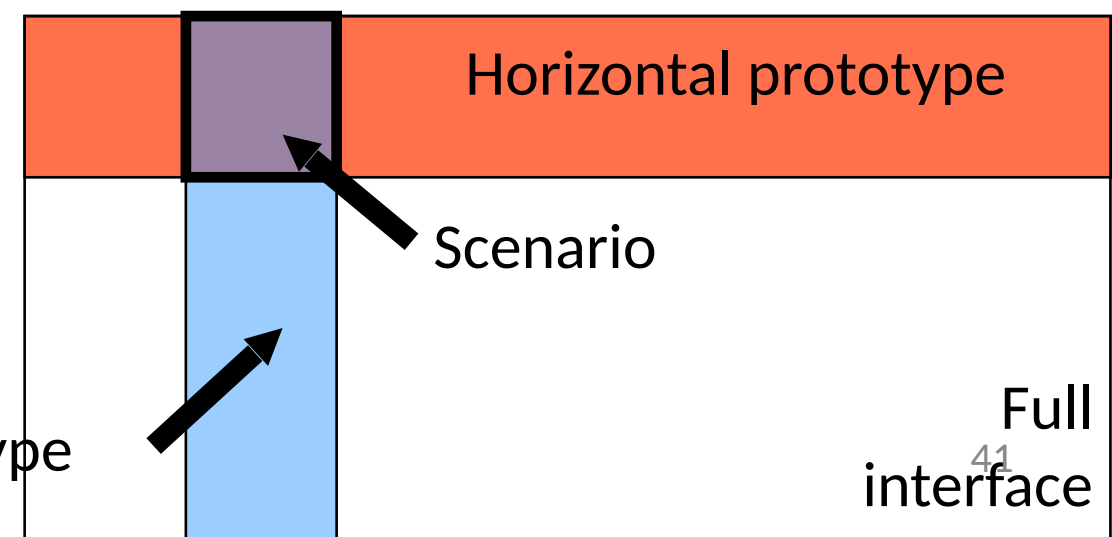
Tasks

- Create functionality necessary for a single task.
- Ex: adding and editing an image

Scenario

- Create functionality needed to run a scenario
- Ex: edit three images and spell-check a document

Vertical prototype



Wireframing

Two dimensional illustration of a page's interface that specifically focuses on

- space allocation
- prioritisation of content
- Functionalities available
- Intended behaviours



Wireframing

What's wrong?

- Focus on screen layout, not interaction
- Mostly static
- Assume buttons, sliders, pulldown menus

You should focus on **interaction**

- Create simple but powerful interaction

scenarios

Like a tiny, breaching one-act play

- Sub-divided into one-paragraph micro scenes that describe a series of 'interaction points'

Create

- one or more personas
- one or more realistic setting(s)

Identify a series of events over a period of time

Balance both

- Normal and unusual situations

Use vs. Design scenarios

Tell a story that illustrates how one or more people interact with technology in a real world setting

Use scenario

- Draws from real-world observation of people who face challenges that a new technology might address

Design scenario

- Builds upon current scenarios and speculates how these people would interact with new technology, in this setting

Change use scenario if it helps you explore alternatives

Design vs. Concept scenarios

Design scenario

- Help interaction designers explore possibilities
- Value: realism, grounded, challenge ideas

Concept and marketing videos

- Help market or 'sell' the concept
- Value: idealised use, market potential

Change use scenario if it helps you explore alternatives

Design Scenarios -> Video prototypes

Design scenario lead to storyboards which lead to video prototypes

Each provides a successively deeper way to think about situated interaction

Each should be considered highly interactive

- Creating alternatives is cheap
- Use them to explore alternatives
- Don't be afraid to try and reject ideas

Video prototypes

Find a quiet place

Bring paper prototypes

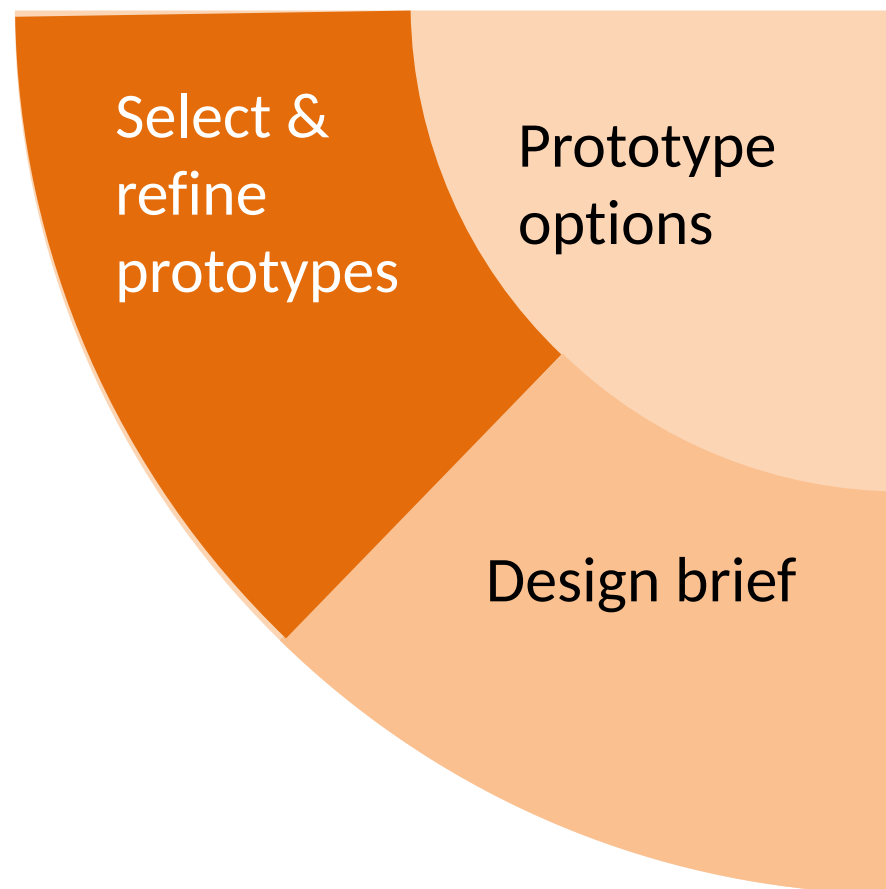
Shoot clips into the storyboard

- If you make a mistake reshoot

Design Resources

- Low-fidelity prototype
- Medium-fidelity prototype
- High-fidelity prototype

Analyze information



Collect information

- sketches

What should it be?

Design

Tips

Focus on interaction, not on interface, functionality

Create a concept

- Identify a real, specific problem rather than a stereotypical 'toy' problem

Trade-off between power and simplicity

- Shift the curve

Be curious, creative, seek surprises and new opportunities

Find an object of interest for the user

- Make it persist, interactive
- Example of a road...

Avoid analysis paralysis

- Choose something!! The first idea is NEVER complete, right or good

Questions?



Questions?

Why should we use paper?

- Sketches and prototypes
- Quick and cheap
- Easy to communicate
- Support evaluation



Questions?

Why should we use paper?

- Sketches and prototypes
- Quick and cheap
- Easy to communicate
- Support evaluation

What should ideally a prototype do?

- It should provide different levels of details
- It should be interactive
- It should allow scenarios testing

